

RISK SHARING IN PRIVATE INFORMATION MODELS WITH ASSET ACCUMULATION: EXPLAINING THE EXCESS SMOOTHNESS OF CONSUMPTION

BY ORAZIO P. ATTANASIO AND NICOLA PAVONI¹

We study testable implications for the dynamics of consumption and income of models in which first-best allocations are not achieved because of a moral hazard problem with hidden saving. We show that in this environment, agents typically achieve more insurance than that obtained under self-insurance with a single asset. Consumption allocations exhibit “excess smoothness,” as found and defined by Campbell and Deaton (1989). We argue that excess smoothness, in this context, is equivalent to a violation of the intertemporal budget constraint considered in a Bewley economy (with a single asset). We also show parameterizations of our model in which we can obtain a closed-form solution for the efficient insurance contract and where the excess smoothness parameter has a structural interpretation in terms of the severity of the moral hazard problem. We present tests of excess smoothness, applied to U.K. microdata and constructed using techniques proposed by Hansen, Roberds, and Sargent (1991) to test the intertemporal budget constraint. Our theoretical model leads us to interpret them as tests of the market structure faced by economic agents. We also construct a test based on the dynamics of the cross-sectional variances of consumption and income that is, in a precise sense, complementary to that based on Hansen, Roberds, and Sargent (1991) and that allows us to estimate the same structural parameter. The results we report are consistent with the implications of the model and are internally coherent.

KEYWORDS: Consumption, private information, hidden savings, excess smoothness.

1. INTRODUCTION

IN THIS PAPER, we study the intertemporal allocation of consumption in a model with moral hazard and hidden saving. We characterize some of the empirical implications of such a model and show how it can explain some well known puzzles in the consumption literature.

The ability to smooth out income shocks and avoid large changes in consumption is an important determinant of individual households' welfare. Resources can be moved over time and across states of the world using a variety of instruments, ranging from saving and borrowing to formal and informal insurance arrangements. The study of the intertemporal allocation of consumption is equivalent to establishing what instruments are available to individual households (that is, what are the components of the intertemporal budget constraint) and how they are used.

¹We are grateful for comments received from Marco Bassetto, Richard Blundell, Harald Uhlig, three anonymous referees, participants at the 2004 SED annual meeting, and several seminar audiences. Attanasio's research was financed by the ESRC Professorial Fellowship Grant RES-051-27-0135. Nicola Pavoni thanks the Spanish Ministry of Science and Technology for Grant BEC2001-1653.

When thinking of the smoothing of individual shocks, two mechanisms come immediately to mind. On one hand, households can use a private technology to move resources over time, for instance, saving in “good times” and dis-saving in “bad times.” On the other, they can enter implicit or explicit insurance arrangements against idiosyncratic shocks. There might be links between these two mechanisms: with endogenously incomplete markets, the availability of specific instruments to move resources over time is likely to affect the type of insurance arrangements that can be sustained in equilibrium. We study consumption smoothing within a unifying framework and consider explicitly the links between private (anonymous) savings and insurance in a moral hazard model. We therefore contribute to and bring together two strands of the literature: that on consumption smoothing and that on endogenously incomplete markets. In particular, we show how some empirical results in the consumption literature can be interpreted, with the help of the model we construct, as providing evidence on partial insurance and on the relevance of specific imperfections.

The permanent income/life-cycle (PILC) model has been, for a long time, the workhorse to study individual consumption smoothing and the intertemporal allocation of resources. During the 1980s and 1990s, a number of papers, starting with Campbell and Deaton (1989), pointed to the fact that consumption seems “too smooth” to be consistent with the model’s predictions, in that consumption does not react sufficiently to innovation to the permanent component of income. This evidence, derived from aggregate data, was interpreted as a failure of the PILC model.

In such a model, allocations are determined by a sequence of Euler equations and an intertemporal budget constraint (IBC), often characterized by the presence of a single asset with a fixed rate of return. We refer to the latter as the risk-free IBC. Campbell and Deaton (1989), as other papers in this literature (including West (1988) and Gali (1991)), took the risk-free IBC (and in particular, the instruments available to an individual to move resources over time) as given and therefore interpreted the evidence of “excess smoothness” as a failure of some of the Euler equations. Indeed, Campbell and Deaton (1989, p. 372) concluded their paper by saying, “Whatever it is that causes changes in consumption to be correlated with lagged changes in income whether [...] the marginal utility of consumption depends on other variables besides consumption, or that consumers are liquidity constrained or that consumers adjust slowly through inertia or habit formation, the same failure is responsible for the smoothness of consumption ...” All these explanations, however, maintain a risk-free IBC in which individuals move resources over time through a single asset with a fixed interest rate. We propose a model in which excess smoothness is related to a violation of the risk-free IBC which arises because the standard model neglects trades in state-contingent assets (or state-contingent transfers) that give individuals more insurance than they would be

able to get by self-insurance and saving. In our model, the amount of risk sharing enjoyed by individuals is determined endogenously and depends on the severity of the moral hazard problem.

As we hint above, an IBC defines the structure of the market to which individuals have access. Therefore, a violation of a specific IBC can be interpreted as a violation of a given market structure. Given that our approach focuses on violation of a risk-free IBC, we use a test of an IBC proposed by Hansen, Roberds, and Sargent (1991; henceforth HRS). The HRS test is particularly appropriate in our context as it can be explicitly interpreted as a test of a market structure. Moreover, because of the way it is formulated, not only does it encompass other tests (such as the Campbell and Deaton (1989) or West (1988)), but it is also robust to informational advantages of the agents relative to the econometrician. The cost of applying the HRS test is that it requires the Euler equation to hold. However, this is consistent with the model we construct and, more importantly, is an empirically testable proposition.

When framed within the context of endogenously incomplete markets, it seems intuitive that a model that predicts some additional insurance relative to the amount of (self-) insurance one observes in a Bewley model would predict that consumption reacts less to shocks than is predicted by the Bewley model. To make such a statement precise, however, is not a trivial exercise. In what follows, in addition to such a formal statement, we also obtain, for some versions of our model, closed-form solutions for the optimal intertemporal allocations, and we map the coefficient of excess smoothness of the consumption literature and of the HRS test into a structural parameter of our model that reflects the severity of the moral hazard problem.

This paper, therefore, makes two main contributions. The first is of a theoretical nature: we show how a model with moral hazard and hidden saving, by providing additional insurance relative to a Bewley economy, generates what in the consumption literature has been called the excess smoothness of consumption. In addition, we construct specific examples in which we can derive a closed-form solution and relate directly the excess-smoothness coefficients to a structural parameter of our model. While the examples are surely special, we stress that, in equilibrium, they deliver an income process that coincides with the process typically used in the permanent income hypothesis (PIH) literature. Moreover, the examples give an intuition that can be generalized to a larger class of models. Our theoretical result allows us to interpret in a specific fashion the empirical findings we obtain. The second contribution is empirical: we adapt the HRS test to microdata and apply it on a time series of cross sections from the United Kingdom. This evidence constitutes one of the first examples of an excess smoothness test performed on microdata. We also complement the HRS test with a test based on movements in the cross-sectional variance of consumption. Because of the nature of our data, we are forced to use the HRS test on time series of cross sections and, therefore, focus on risk sharing across cohorts. The variance test complements the HRS test because it focuses on insurance within cohorts.

Our empirical results, obtained from U.K. time series of cross sections, are remarkably in line with the predictions of the model we construct.

The rest of the paper is organized as follows. In Section 2, we present the building blocks of our model, provide the equilibrium definition, and discuss alternative market structures. In Section 3, we characterize the equilibrium allocation for our model and present some examples that yield useful closed-form solutions. In Section 4, we discuss the empirical implications of the equilibria we considered in Section 3, and present our tests and the results we obtain. Section 5 concludes. The Appendices contain the proofs of the results stated in the text: Appendices B and C comprise the Supplemental Material (Attanasio and Pavoni (2011)).

2. MODEL

2.1. *Tastes and Technology*

Consider an economy consisting of a large number of agents that are ex ante identical. Each agent lives $T \leq \infty$ periods and is endowed with a private stochastic production technology which takes the form (neglecting individual indices for notational ease)

$$y_t = f(\theta_t, e_t).$$

That is, the individual income $y_t \in Y \subset \Re$ is determined by the agent's effort level $e_t \in E \subset \Re$ and the shock $\theta_t \in \Theta \subset \Re$. The history of income up to period t is denoted by $y^t = (y_1, \dots, y_t)$, while the history of shocks is $\theta^t = (\theta_1, \dots, \theta_t)$. Let $\Phi(\theta_{t+1}|\theta^t)$ be the conditional probability of θ_{t+1} conditioned on $\theta^t \in \Theta^t$. The component θ_t can be interpreted as the agent's skill level at date t . At this stage, we do not impose any specific structure on the time series properties of θ_t , but we assume that θ^t are independent and identically distributed (i.i.d.) across individuals. In each period, the effort e_t is chosen after observing θ_t . The function $f: \Theta \times E \rightarrow Y$ is assumed to be continuous and increasing in both arguments. Both the effort e and the shocks θ are assumed to be private information, giving rise to moral hazard problems, while y_t is publicly observable.

Agents are born with no wealth, have von Neumann–Morgenstern preferences, and rank deterministic sequences according to

$$\sum_{t=1}^T \delta^{t-1} u(c_t, e_t),$$

with $c_t \in C$ and $\delta \in (0, 1)$. We assume u to be real valued, continuous, strictly concave, smooth, strictly increasing in c_t , and strictly decreasing in e_t . Notice that, given a plan for effort levels, there is a deterministic and one-to-one mapping between histories of the private shocks θ^t and income y^t . Therefore, we

can consider θ^t alone. Let μ^t be the probability measure on Θ^t and assume that the law of large numbers applies, so that $\mu^t(\mathcal{A})$ is also the fraction of agents with histories $\theta^t \in \mathcal{A}$ at time t .

Since θ^t are unobservable, we make use of the revelation principle and define a reporting strategy $\sigma = \{\sigma_t\}_{t=1}^T$ as a sequence of θ^t -measurable functions such that $\sigma_t: \Theta^t \rightarrow \Theta$ and $\sigma_t(\theta^t) = \hat{\theta}_t$ for some $\hat{\theta}_t \in \Theta$. A truthful reporting strategy σ^* is such that $\sigma_t^*(\theta^t) = \theta_t$ almost surely (a.s.) for all θ_t . Let Σ be the set of all possible reporting strategies. A reporting strategy essentially generates publicly observable histories according to $h^t = \sigma(\theta^t) = (\sigma_1(\theta_1), \dots, \sigma_t(\theta^t))$, with $h^t = \theta^t$ when $\sigma = \sigma^*$.

An allocation (e, c, y) consists of a triplet $\{e_t, c_t, y_t\}_{t=1}^T$ of θ^t -measurable functions for effort, consumption, and income growths (production) such that they are “technically” attainable:

$$\Omega = \{(e, c, y) : \forall t \geq 1, \theta^t, e_t(\theta^t) \in E, c_t(\theta^t) \in C, \text{ and} \\ y_t(\theta^t) = f(\theta_t, e_t(\theta^t))\}.$$

The idea behind this notation is that incentive compatibility will guarantee that the agent announces truthfully his endowments (i.e., uses σ^*) so that, in equilibrium, private histories are public information.

For simplicity, we disregard aggregate shocks and we do not allow for productive assets such as capital. In the baseline model, we assume the availability of a constant return technology that allows $q \in (0, 1)$ units of consumption at time t to be transformed into one unit of consumption at time $t + 1$ or vice versa. Equivalently, we are assuming a small open economy or a village economy where a money lender has access to an external market with an exogenously given interest rate r . The number $q = \frac{1}{1+r}$ can be interpreted as the time constant price of a one-period bond in the credit market. In this case, absent aggregate shocks, thanks to the law of large numbers, the feasibility condition is constituted to be a unique inequality:

$$(1) \quad \int_{\Theta^t} \left[\sum_{t=1}^T q^{t-1} c_t(\theta^t) \right] d\mu^t(\theta^t) \leq \int_{\Theta^t} \left[\sum_{t=1}^T q^{t-1} y_t(\theta^t) \right] d\mu^t(\theta^t).$$

Although we present our results under the assumption of a small open economy with a constant interest rate, in the Appendices we show that the same results can be derived with a time-varying interest rate and even for the case of a closed economy.²

²For a similar model in a small open economy, see Abraham and Pavoni (2004, 2005, 2008). For another similar analysis in a closed economy with capital, see Golosov and Tsyvinski (2007).

2.2. Equilibrium and Market Arrangements

Having described agents' tastes and the technological environment they face, to characterize intertemporal allocations we need to specify the market arrangements in which they operate and define the relevant equilibrium concepts.

Our economy is characterized by private information, as we assume that effort and skill level are not observable. In addition to the moral hazard problem, consumption is also not observable (and/or contractable) and agents have hidden access to an anonymous credit market where they trade risk-free bonds at price q . The agents do not have private access to any other anonymous asset markets.³

The market structure we adopt is an open economy version of that proposed in Golosov and Tsyvinski (2007). In addition to the anonymous (secondary) market for assets, there is a (primary) insurance market where a continuum of identical firms offer exclusive contracts to agents. All insurance firms are owned equally by all agents. At the beginning of period 1, each firm signs a contract (e, c, y) with a continuum of agents in the economy that is binding for both parties. Firms operate in a competitive market and agents sign a contract with the firm that promises the highest ex ante expected discounted utility. After the contract is signed, each agent chooses a reporting strategy σ , supplies effort levels so as to generate, at each node, the agreed income level $y_t(\sigma(\theta^t))$, and receives $c_t(\sigma(\theta^t))$ units of consumption. Firms can borrow and lend at the ongoing interest rate, which they take as given.

We define expected utility from reporting strategy $\sigma \in \Sigma$, given the allocation $(e, c, y) \in \Omega$, as

$$\begin{aligned} & \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(c_t, e_t) \middle| (e, c, y), \sigma \right] \\ & := \sum_{t=1}^T \delta^{t-1} \int_{\theta^t} u(c_t(\sigma(\theta^t)), g(\theta_t, y_t(\sigma(\theta^t)))) d\mu^t(\theta^t), \end{aligned}$$

where $g(y, \theta)$ represents the effort level needed to generate y when the shock is θ , that is, g is the inverse of f with respect to e keeping θ fixed. Since y is observable, the misreporting agent must adjust his/her effort level so that the lie is not detected.

Let $b := \{\mathbf{b}_{t+1}\}_{t=1}^T$ be a plan of risk-free asset holding, where $\mathbf{b}_t$ is a θ^{t-1} -measurable function with $b_1 = 0$ and $\lim_{t \rightarrow T} q^{t-1} \mathbf{b}_{t+1}(\theta^t) \geq 0$. Given the price

³The assumption that in the anonymous asset market only the risk-free asset is traded is done without loss of generality in our environment, as no other Arrow security would be traded (see Golosov and Tsyvinski (2007)).

of the bond q , the typical firm, in equilibrium, solves the problem

$$(2) \quad \max_{b, (e, c, y) \in \Omega} \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(c_t, e_t) \middle| (e, c, y), \sigma^* \right]$$

$$(3) \quad \text{subject to } \mathbf{E} \left[\sum_{t=1}^T q^{t-1} (c_t - y_t) \middle| (e, c, y), \sigma^* \right] \leq 0,$$

with the incentive compatibility constraint

$$(4) \quad \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(c_t, e_t) \middle| (e, c, y), \sigma^* \right] \\ \geq \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(\hat{c}_t^\sigma, \hat{e}_t^\sigma) \middle| (e, c, y), \sigma \right] \quad \text{for all } \sigma \in \Sigma,$$

where the deviation for consumption $\hat{c}^\sigma$ must be such that the new path of consumption can be replicated by the use of a risk-free bond. More precisely, given (e, c, y, b) , for each σ , a deviation $\hat{c}^\sigma$ is admissible if there is a plan of bond holdings $\hat{b}^\sigma$ such that for all t and a.s. for all histories θ^t ,

$$\begin{aligned} \hat{c}_t^\sigma(\sigma_t(\theta^t)) + q\hat{b}_{t+1}^\sigma(\sigma_t(\theta^t)) - \hat{b}_t^\sigma(\sigma_{t-1}(\theta^{t-1})) \\ = c_t(\sigma_t(\theta^t)) + qb_{t+1}(\sigma_t(\theta^t)) - b_t(\sigma_{t-1}(\theta^{t-1})) \end{aligned}$$

and $\lim_{t \rightarrow T} q^{t-1} \hat{b}_{t+1}^\sigma(\sigma_t(\theta^t)) \geq 0$.

DEFINITION 1: Given q , an equilibrium for the economy is an allocation (e^*, c^*, y^*) and bond trades b^* such that the following statements hold:

- (i) Firms choose (e^*, c^*, y^*, b^*) , solving problem (2)–(4) (taking q as given).
- (ii) Agents choose their reporting strategy, effort levels, consumption, and asset trades optimally as described above, given the contract (and q).
- (iii) The intertemporal aggregate feasibility constraint (1) holds.

It is straightforward to see that the incentive constraint (4) (considered at σ^*) implies that the equilibrium allocation must satisfy the Euler equation

$$(5) \quad u'_c(c_t^*, e_t^*) = \frac{\delta}{q} \mathbf{E}_t[u'_c(c_{t+1}^*, e_{t+1}^*)],$$

where the marginal utilities are evaluated at the equilibrium values dictated by (c^*, c^*, y^*) and $\mathbf{E}_t[\cdot]$ is the conditional expectation operator on future histories given θ^t .⁴

2.3. Two Useful Extreme Cases

With an eye to our empirical strategy, it is useful to compare the allocations in our imperfect information model with those that obtain in two well known alternative and extreme cases: a full information setting and a situation where intertemporal trades are exogenously restricted to the risk-free bond. Both situations can be obtained, under certain parameter settings, as special cases of our model. Moreover, both can be characterized by a different intertemporal budget constraint.

Full Information

In the model where there is no private information problem, in equilibrium, the typical firm offers exclusive contracts solving problem (2) and (3) alone. That is, the firm will solve the same problem as before with the crucial exception of condition (4). Since in this case we are in a complete market setting, we can apply the well known welfare theorems which imply that the equilibrium allocation solves the problem of a planner aiming to maximize the representative agent's utility subject to the feasibility constraint, condition (1).

Note that, as in our setup, with full information the equilibrium allocation satisfies the Euler equation (5). In fact, the equality between present and future discounted marginal utility is true state by state, $u'_c(c_t, e_t) = \frac{\delta}{q} u'_c(c_{t+1}, e_{t+1})$, and hence also in expected terms.

The Bewley Economy (Self-Insurance)

We define a Bewley economy (or self-insurance) as an economy where agents can move resources over time only by participating in a simple credit market with a constant one-period bond with price q . Recall that an asset holding plan b is made of a set of θ^{t-1} -measurable function $\mathbf{b}_t$, $t = 1, \dots, T$, with $b_1 = 0$. In this economy, a typical agent solves

$$\max_{b, (e, c, y) \in \Omega} \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(c_t, e_t) \right]$$

subject to

$$(6) \quad \mathbf{c}_t(\theta^t) + q\mathbf{b}_{t+1}(\theta^t) \leq \mathbf{b}_t(\theta^{t-1}) + \mathbf{y}_t(\theta^t),$$

⁴This condition is the first-order equivalent of the incentive constraint that ensures that the agent is not willing to deviate in assets decisions alone, while contemplating telling the truth about shock histories θ^t .

where $b_0 = 0$. As usual, we rule out Ponzi games by requiring that $\lim_{t \rightarrow T} q^{t-1} \times \mathbf{b}_t(\theta^t) \geq 0$. Condition (6) is the budget constraint typically used in permanent income models when the agent has access only to a risk-free bond market. This problem can be seen as an extension of the permanent-income model studied by Bewley (1977), which allows for endogenous labor supply and nonstationary income.

The main implication of the self-insurance model is the well known Euler equation

$$(7) \quad u'_c(c_t, e_t) = \frac{\delta}{q} \mathbf{E}_t[u'_c(c_{t+1}, e_{t+1})].$$

Another important necessary condition that individual intertemporal allocations have to satisfy in this model can be derived by repeatedly applying the budget constraint (6). Starting from any node $\bar{\theta}^{t-1}$, $t \geq 1$, with asset holding level $\mathbf{b}_t(\bar{\theta}^{t-1})$, the following net present value condition (NPVC) must be satisfied:

$$(8) \quad \sum_{n=t}^T q^{n-t} (\mathbf{c}_n(\theta^n) - \mathbf{y}_n(\theta^n)) \leq \mathbf{b}_t(\bar{\theta}^{t-1})$$

a.s. for all histories θ^T emanating from node $\bar{\theta}^{t-1}$.

Given the income process and the price for the bond q , conditions (7) and (8) define consumption (even when a closed-form solution does not exist). It is interesting to compare NPVC (8) for $t = 1$ with the corresponding equation for the full information case, equation (1). In the latter case, the agent has available a wide array of state-contingent securities that are linked in an individual budget constraint that sums over time and across histories, as all trades can be made at time 1. In the Bewley economy, instead, the agent can only trade in a single asset. This restriction on trade requires that the net present value of consumption minus income equals zero a.s. for *all* histories θ^T . Notice that there is no expectation operator involved in condition (8): the intertemporal transfer technology implied by a risk-free asset does not allow for cross-subsidizations of consumption across income histories.

The lesson we should retain from this section is that all allocations considered here satisfy the Euler equation: clearly such an equation is consistent with many stochastic processes for consumption. The key feature that distinguishes the three different equilibrium allocations we have considered is the intertemporal budget constraint, which defines the relevant market structure. For instance, the equilibrium allocation in our moral hazard model typically violates the NPVC (8) based on a *single asset* which is relevant for the permanent income model because it ignores the state-contingent payments the agents might receive from the insurance firm in our economy.

3. CHARACTERIZING THE EQUILIBRIUM ALLOCATION

In this section, we analyze the properties of the endogenously incomplete markets model we presented above. It is easy to show that the equilibrium allocation (e^*, c^*, y^*) and b^* can be replicated by an incentive compatible plan of lump sum transfers $\tau^* = \{\tau_t^*(\theta^t)\}_{t=1}^T$ that solves the firm's problem

$$\max_{\tau, (e, c, y) \in \Omega} \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(c_t, e_t) \right]$$

subject to

$$(9) \quad c_t(\theta^t) = y_t(\theta^t) + \tau_t(\theta^t),$$

the incentive constraint (4), and the intertemporal budget constraint

$$\mathbf{E} \left[\sum_{t=1}^T q^{t-1} \tau_t \right] \leq 0.$$

It is clear from condition (9) that we study equilibrium allocations where agents do not trade intertemporally ($b_t^* \equiv 0$). This is done without loss of generality since the firms and the agents face the same interest rate.⁵ Alternatively, τ could be chosen so that the transfer $\tau_t = \tau_t(\theta^t)$ represents the net trade on state-contingent assets the agent implements at each date t and node θ^t . In this case, the lump sum transfer solves $\mathbf{E}[\tau_t] = 0$ and all the intertemporal transfer of resources are, in effect, made by the agents.

In Appendix A, we show that, under some conditions, the equilibrium allocation (e^*, c^*, y^*) and $b_t^* \equiv 0$ can be implemented using a transfer scheme τ^* which is a function of income histories y^t alone.⁶ This simplifies the analysis and allows us to describe the consumption allocation in terms of observables. At history y^t , an agent with asset level b_t faces the following budget constraint for all y^t :

$$c_t + qb_{t+1} = y_t + \tau_t^*(y^t) + b_t.$$

3.1. Excess Smoothness and the Bewley Economy

Allen (1985) and Cole and Kocherlakota (2001) presented asymmetric information models that give rise to intertemporal allocations that coincide with those that prevail in the Bewley economy we described in Section 2.3. In this section, we first present a specific version of our model where the Allen and

⁵For example, see Fudenberg, Holmstrom, and Milgrom (1990).

⁶In particular, we assume that the optimal plan of consumption c is y^t -measurable.

Cole–Kocherlakota (ACK) result obtains. As stressed in Abraham and Pavoni (2004), to obtain the self-insurance result, a crucial restriction on the way effort is converted into output is needed. We then move on to relax this restriction. Within the more general case, we consider a specific parameterization of the income process that allows us to obtain a closed-form solution for the equilibrium transfers. While this example is useful because it gives very sharp predictions, some of the properties of the allocations we discuss generalize to the more general case and inform our empirical specification.

Self-Insurance

Let us assume agents' preferences and production satisfy

$$(10) \quad u(c, e) = u(c - e) \quad \text{and} \quad f(\theta, e) = \theta + e,$$

with $\Theta = (\theta_{\min}, \theta_{\max})$ and $E = (e_{\min}, e_{\max})$. Obviously, in this environment the optimal plan of effort levels is indeterminate. As long as $0 \in E$, we can hence set, without any loss of generality, $e_t^* \equiv 0$. This normalization has two advantages. First, since e_t does not change with θ_t while $f(\theta_t, e_t)$ is strictly increasing in θ_t , all variations in θ_t will induce variations in y_t , automatically guaranteeing the y^t -measurability of c (see Appendix A). Second, with constant effort we can focus on the risk sharing dimension of the equilibrium allocation. This last argument also motivates the modelling choice for our closed-form solution below.

PROPOSITION 1: *Assume $T < \infty$, and that the utility function u and the production function f are as in (10). Then the equilibrium allocation coincides with self-insurance.*

The proof of the proposition is reported in Appendix A, where we show that incentive compatibility fully characterizes the equilibrium allocation. The main intuition of the result can be easily seen in a two-period model, which we solve backward. Consider the last period of the program. Using the budget constraint, given any history y^{T-1} and θ_T (and last period wealth b_T), the incentive constraint (for $e_T^* = 0$) is

$$\begin{aligned} u(c_T^* - e_T^*) &= u(\theta_T + \tau_T^*(y^{T-1}, \theta_T) + b_T) \\ &\geq u(\theta_T + \tau_T^*(y^{T-1}, \theta_T + \hat{e}_T) + b_T) \quad \text{for all } \hat{e}_T \in E. \end{aligned}$$

Clearly, to be incentive compatible, the transfer scheme must be constant across y_T s. Now consider the problem in period $T - 1$. For ease of exposition, let us assume that the transfer scheme is differentiable⁷ and write the

⁷The formal proof does not assume differentiability. See Appendix A.

agent's first-order conditions with respect to e_{T-1} (evaluated at the optimum $b_T^* = b_{T-1}^* = e_T^* = e_{T-1}^* = 0$). A necessary condition for incentive compatibility is⁸

$$\frac{\partial \tau_{T-1}^*(y^{T-1})}{\partial y_{T-1}} + \delta \frac{\partial \tau_T^*(y^T)}{\partial y_{T-1}} \mathbf{E}_{T-1} \left[\frac{u'(c_T^*)}{u'(c_{T-1}^*)} \right] = 0.$$

The Euler equation—that is, the (local) incentive constraint for bond holding—reads as $\mathbf{E}_{T-1} \left[\frac{u'(c_T^*)}{u'(c_{T-1}^*)} \right] = \frac{q}{\delta}$, which implies

$$(11) \quad \frac{\partial \tau_{T-1}^*(y^{T-1})}{\partial y_{T-1}} + q \frac{\partial \tau_T^*(y^T)}{\partial y_{T-1}} = 0.$$

Equation (11) states that the net present value of transfers must be constant across income histories. The intuition for the fact that the firm is unable to provide any consumption insurance on top of self-insurance is relatively simple in this case. First, at each t with equilibrium income $y_t^* = \theta_t$, the agent can always deviate, locally, and generate any income level such that $y_t^* - \varepsilon \leq \hat{y}_t \leq y_t^* + \varepsilon$. Second, the perfect substitutability between consumption and effort in the utility function on one side, and between income and effort in production on the other side imply that such deviation has *zero direct cost* to the agent. Since this is true for all income levels, in a static setting, these two observations together imply that the firm will never be able to provide a transfer scheme that induces anything different from constant payments over income levels. In our model, this simple intuition extends to a general dynamic setting. Roughly speaking, free access to the credit market implies that the agent only cares about the *present value* of transfers (i.e., he/she does not care about the exact timing of deterministic transfer payments). Hence, a simple extension of the previous argument implies that the present value of transfers must be constant across income histories, as otherwise the agent would find it profitable to perform some local deviation on effort and engage in an appropriate bond plan.

Now recall that the self-insurance allocation has two defining properties: first, it must satisfy the Euler equation; second, it must satisfy the intertemporal budget constraint with one bond, that is, the period zero net present value must be zero for all y^T . Since the Euler equation is always satisfied here, the only way to obtain a different allocation is that the transfers scheme τ^* permits violation of the agent's period zero self-insurance intertemporal budget constraint for some history y^T . The previous argument demonstrates that this cannot be the case; hence the only incentive compatible allocation coincides with self-insurance. This implies that the “relaxed-optimal” contract obtained

⁸Note that in the expression below, $\frac{\partial \tau_T^*(y^T)}{\partial y_{T-1}}$ has been taken out from the expectation operator as we saw that τ_T is constant in y_T shocks.

by using the local (first-order-condition) version of the incentive constraint corresponds to the self-insurance allocation. Since this allocation is obviously globally incentive compatible, it must be the optimal one.

The proof of Proposition 1, in Appendix A, never uses the time-series properties of θ_t , which can, therefore, be very general. Likewise, we do not require a constant q .⁹ Proposition 1 is, therefore, in some dimensions, a general result. Indeed, as we prove in Appendix B, it holds even in a more general model that allows for two types of income shocks, with different degrees of persistence.

Proposition 1 and its proof, however, also make it clear that the coincidence between the equilibrium allocation and self-insurance is, in other dimensions, very fragile. Below, we present a full class of models with slightly more general u and f functions, where the equilibrium allocation coincides with self-insurance only for a zero measure set of parameters.¹⁰

Excess Smoothness

Consider now the generalization of agent preferences

$$u(c, e) = u(c - v(e)),$$

with v increasing and convex. Since consumption and *effort cost* enter the utility function in a linear fashion, we eliminate the wealth effects. This simplifies the analysis and allows for closed-form solutions. Moreover, it is crucial for the self-insurance result we derived above. While we only consider interior solutions for e , we leave the function f unspecified.

Again, the main intuition regarding the model can be gained by studying the last two periods. Let us start with the final period (T). The first-order condition of the incentive constraint in this last period is

$$(12) \quad 1 + \frac{\partial \tau_T^*(y^{T-1}, y_T)}{\partial y_T} = \frac{v'(e_T^*)}{f'_e(e_T^*, \theta_T)}.$$

Recall that in the ACK model we have $v'(e) = f'_e(e, \theta) = 1$ for all e, θ . Since risk sharing requires $\frac{\partial \tau_T^*(y^{T-1}, y_T)}{\partial y_T} < 0$, no insurance is possible in that environment. However, in general, $\frac{v'(e_T)}{f'_e(e_T, \theta_T)}$ might be less than 1. In fact, it is easy to see that under some regularity conditions on f and v , for all θ_T , the optimal

⁹Both Allen (1985) and Cole and Kocherlakota (2001) assumed i.i.d. shocks and constant q . We generalize Allen's result, while our model does not nest, strictly speaking, Cole and Kocherlakota's, as we do not impose exogenous and binding liquidity constraints.

¹⁰Abraham and Pavoni (2004) also obtained similar results. They considered a more general class of models but did not derive sufficient conditions for the self-insurance result and did not obtain closed-form solutions.

contract implements effort levels such that $\frac{v'(e_T)}{f'_e(e_T, \theta_T)} \leq 1$ (as long as this is technologically feasible).¹¹ A strict inequality is compatible with some risk sharing.

What is the intuition for this fact? If the firm's aim is to make agents share risk, the key margin for an optimal contract is to guarantee that the agent does not shirk or, equivalently, that she does not *reduce* effort. When the agent shirks so that output is reduced by one unit, the gain she gets from this reduction in effort is equivalent to $\frac{v'}{f'_e}$ units of consumption. This is the right-hand side of (12). The left-hand side is the net consumption loss. When the marginal tax/transfer is negative, this loss will be less than 1 as the direct reduction of one unit of consumption is mitigated by the increase in net transfers. A small $\frac{v'}{f'_e}$ reduces shirking returns, making it easier for the firm to satisfy the incentive compatibility and hence to provide insurance. The same intuition carries over to a multiperiod setting, where the generalized version of (11) takes the form

$$\frac{\partial \tau_{T-1}^*(y^{T-1})}{\partial y_{T-1}} + q \frac{\partial \tau_T^*(y^T)}{\partial y_{T-1}} = \frac{v'(e_{T-1}^*)}{f'_e(\theta_{T-1}, e_{T-1}^*)} - 1.$$

3.2. Closed Forms

We now assume that income y_t depends on exogenous shocks θ_t and effort e_t as

$$(13) \quad y_t = f(\theta_t, e_t) = \theta_t + a \min\{e_t, 0\} + b \max\{e_t, 0\},$$

with $a \geq 1 \geq b$. Notice that when $a = b = 1$, one obtains the linear specification used to obtain the self-insurance result. Preferences are as in the previous section¹²:

$$u(c_t, e_t) = u(c_t - e_t).$$

Moreover, we assume that θ_t follows an ARIMA(p) process

$$(14) \quad \theta_t - \theta_{t-1} = \beta(L)v_t,$$

where $\beta(L)$ is a polynomial of order p in the lag operator L , invertible, and the innovation v_t is i.i.d. Linearity of the autoregressive integrated moving average

¹¹Assume there is at least one $\bar{e} \in E$ such that $\frac{v'(\bar{e})}{f'_e(\bar{e}, \theta)} \leq 1$. Implementing an effort level where $\frac{v'(e_T)}{f'_e(e_T, \theta_T)} > 1$ is dominated by a lower effort, since by marginally reducing e_T , for such θ_T , there is a production gain. In the static case, it can be shown that if $\frac{d}{de}[\frac{f'_\theta v'}{f'_e}] \geq 0$ (i.e., the Spence–Mirrlees condition holds for the static version of our problem), the firm can provide the same ex post welfare to the agent by saving net transfer costs as it also improves on insurance. Details on the formal derivation are available upon request.

¹²Equivalently, we could have assumed a fully linear production function, $f(e, \theta) = e + \theta$, and a kinked effort cost function, $v(e) = \frac{1}{a} \min\{e_t, 0\} + \frac{1}{b} \max\{e_t, 0\}$.

(ARIMA) process helps us to find the closed-form solution. Since effort will be time constant, in equilibrium, $\theta_t = y_t$, and hence the income process will display the standard representation often used in the consumption literature.¹³

In Proposition 3, stated and proved in Appendix B, we show that if u is exponential (constant absolute risk aversion (CARA)), $u(c - e) = -\frac{1}{\rho} \exp\{-\rho(c - e)\}$, and the shocks v_t are *normally distributed* with zero mean and variance σ_v^2 , we get the expression for consumption growth

$$c_t^* - c_{t-1}^* = \frac{\ln(\delta/q)}{\rho} + \frac{\rho}{2} \sigma_c^2 + \frac{1}{a} \frac{(1-q)}{1-q^{T-t+1}} (\mathbf{E}_t - \mathbf{E}_{t-1}) \left[\sum_{n=0}^{T-t} q^n y_t^* \right],$$

where σ_c^2 indicates the variance of consumption growth. For $t \leq T - p$, the previous expression reduces to (see Appendix B for details)

$$(15) \quad \Delta c_t^* = \frac{\ln(\delta/q)}{\rho} + \frac{\rho}{2a^2} [\beta(q)]^2 \sigma_v^2 + \frac{1}{a} \beta(q) v_t.$$

For $a = 1$, we are back to the self-insurance case where innovations to permanent income are fully reflected in consumption changes. For $a > 1$, we get some more risk sharing over and above self-insurance, with full insurance obtainable as a limit case for $a \rightarrow \infty$. Proposition 5 in Appendix B, shows that a very similar (simpler) closed form can be obtained assuming quadratic u , $\delta = q$, and differentiability of the transfer scheme.¹⁴ With CARA utility (as opposed to the model where u is quadratic), the presence of a precautionary saving motive implies that the equilibrium allocation displays increasing consumption. Notice, however, that in our economy, $a > 1$ permits both reduction of the cross-sectional dispersion of consumption and mitigation of the precautionary saving motive, hence the steepness of consumption (i.e., intertemporal dispersion). The model implies a very tight relationship between these two moments.

3.3. The Case With Isoelastic Utility

We now discuss the case in which agents have isoelastic preferences. Full details are to be found in Appendix B. We assume the production function

$$y_t = \theta_t e_t.$$

This model corresponds to a simple dynamic variant of the most standard version of the well known model of Mirrlees (1971), where y is labor income, θ

¹³See, for example, Abowd and Card (1989), Meghir and Pistaferri (2004), and Blundell, Pistaferri, and Preston (2008).

¹⁴In this case, no parametric assumptions are needed for the income process and consumption does not grow because a precautionary saving motive is absent. See Appendix B for details.

represents worker productivity, and e is hours of work. We assume that preferences take the CRRA (or isoelastic) form¹⁵

$$\begin{aligned} u(c_t, e_t) &= \frac{(c_t \cdot e_t^{-1/a})^{1-\gamma}}{1-\gamma} \quad \text{for } e_t \leq 1 \quad \text{and} \\ u(c_t, e_t) &= \frac{(c_t \cdot e_t^{-1/b})^{1-\gamma}}{1-\gamma} \quad \text{for } e_t \geq 1, \quad \text{for } \gamma > 1; \\ u(c_t, e_t) &= \ln c_t - \frac{1}{a} \ln e_t \quad \text{for } e_t \leq 1 \quad \text{and} \\ u(c_t, e_t) &= \ln c_t - \frac{1}{b} \ln e_t \quad \text{for } e_t \geq 1, \quad \text{for } \gamma = 1, \end{aligned}$$

where $a \geq 1 \geq b$. As before, therefore, we assume that the marginal cost of effort changes discontinuously above a threshold level, here $e_t = 1$. Moreover, we assume that

$$\ln \theta_t - \ln \theta_{t-1} = \beta(L) \ln v_t,$$

where (with a small abuse in notation) we posit that $\ln v_t$ is normally distributed with zero mean and variance σ_v^2 . In this context, we can obtain a closed form for discounted net transfers that leads, for $t \leq T - p$, to the permanent income formulation

$$(16) \quad \Delta \ln c_t^* = \frac{\ln(\delta/q)}{\gamma} + \frac{\gamma}{2} \sigma_c^2 + \frac{1}{a} \beta(q\lambda) \ln v_t,$$

where $\lambda = \exp\{\frac{\ln \delta/q}{\gamma} + \frac{1-\gamma}{2} \sigma_c^2\}$ and $\sigma_c^2 = \frac{1}{a^2} [\beta(q\lambda)]^2 \sigma_v^2$.¹⁶ Note that whenever u is logarithmic ($\gamma = 1$), then $q\lambda = \delta$ (see Appendix B for details). When $a = 1$ and income shocks are a random walk, $\beta(q\lambda) = 1$ and one obtains the same expression as in the self-insurance model with zero wealth (e.g., Constantinides and Duffie (1996)). With nonzero wealth, the comparison with the Bewley model is more delicate. First, the coefficient that relates innovations to permanent income to consumption needs to be corrected for the wealth-income ratio, as

¹⁵Again, we could have assumed, equivalently, $u(c_t, e_t) = \frac{(c_t \cdot e_t^{-1})^{1-\gamma}}{1-\gamma}$ and a modified Cobb–Douglas production function of the form $y_t = \theta_t e_t^a$ for $e_t \leq 1$ and $y_t = \theta_t e_t^b$ for $e_t \geq 1$, with $a \geq 1 \geq b$.

¹⁶More precisely, given a, γ, σ_v , and the coefficients $\beta_j, j = 0, 1, 2, \dots, p$, defining the persistence of the income process, the conditional variance of consumptionsolves the system of equations

$$\lambda = \exp\left\{\frac{\ln \delta/q}{\gamma} + \frac{1-\gamma}{2} \sigma_c^2\right\} \quad \text{and} \quad \sigma_c^2 = \frac{\sum_{j=0}^p \beta_j q^j \lambda^j \sigma_v}{a}.$$

pointed out by Banks, Blundell, and Brugiavini (2001) and Blundell, Pistaferri, and Preston (2008). Second, equation (16) does not necessarily hold and log consumption is not necessarily a martingale, and this might have implications for our empirical approach, which we discuss below.

“Pure” Moral Hazard

Finally, we can use equation (16) to draw attention to an important difference between our model and a model with observable assets. It is well known that in that case, when preferences are additive separable, the inverse of the marginal utility follows a martingale and, therefore, the Euler equation is violated (see Rogerson (1985) and Ligon (1998)). In that framework, it is easy to show that with isoelastic utility, consumption growth follows¹⁷

$$(17) \quad \mathbf{E}_{t-1} \Delta \ln c_t^* = \frac{\ln(\delta/q)}{\gamma} - \frac{\gamma}{2} \sigma_c^2.$$

The key difference between this expression and (16) is that, according to (17), consumption growth *decreases* with an increase in the variance of consumption and, therefore, income shocks. This prediction is inconsistent with the evidence in, for instance, Banks, Blundell, and Brugiavini (2001).

4. EMPIRICAL IMPLICATIONS OF THE MODEL

The model we developed in the previous section has some important empirical implications. In it, while the Euler equation holds, the risk-free IBC is violated. This violation gives rise to what the literature has labelled the excess smoothness of consumption: consumption does not react enough to innovations in permanent income. We provide one of the first excess-smoothness tests performed on microdata and we give a structural interpretation to the excess-smoothness coefficient as reflecting the severity of moral hazard.¹⁸

In this section, we pursue these ideas by developing two different and complementary tests of the implications of our model on microdata. The first is an excess-smoothness test derived from a time series model of consumption and income, and based on the test of the risk-free IBC proposed by HRS. As mentioned above, testing the validity of an IBC is equivalent, in our context, to testing a market structure. The second test is based on the dynamics of consumption and income inequality within groups, as measured by variances of log consumption. As we discuss below, the two tests complement each other.

As we study the extent to which income shocks are insured or are reflected in consumption, we need to use individual-level data. Unfortunately, however,

¹⁷With CARA utility, the same expression holds true for consumption in levels.

¹⁸Nalewaik (2006) adapted the HRS test to U.S. microdata, but did not stress the relationship with excess smoothness or provides an excess sensitivity test.

longitudinal surveys that follow individuals over time and contain complete information on consumption are extremely rare. One of the most commonly used panels from the United States, the Panel Study of Income Dynamics (PSID), for instance, only contains information on food consumption. Other data sets, which contain complete information on consumption, have a very short longitudinal dimension (such as the Consumer Expenditure Survey (CEX) in the United States) or lack it completely (such as the Family Expenditure Survey (FES) in the United Kingdom). As a long time period is crucial to identify the time series properties of the variables of interest, this lack of longitudinal data is a problem for us. To overcome this difficulty, we use synthetic cohort data or pseudopanel, along the lines proposed by Deaton (1985) and Browning, Deaton, and Irish (1985).

Our main data source is the U.K. FES from 1974 to 2002. The FES is a time series of repeated cross sections which is collected for the main purpose of computing the weights for the Consumer Price Index. Each survey consists of about 7,000 families contacted over 2-week periods throughout the year. As households are interviewed every week throughout the year, the FES data are used to construct quarterly time series. This allows us to exploit a relatively long time series horizon. We use data on households headed by individuals born in the 1930s, 1940s, 1950s, and 1960s to form pseudopanel for four year-of-birth cohorts. As we truncate the samples so as to have individuals aged between 25 and 60, the four cohorts form an unbalanced sample. The 1930s cohort is observed over later periods of its life cycle and exits before the end of our sample, while the opposite is true for the 1960s cohort. Apart from the year of birth, the other selection criterion we used for this study is marital status. As we want to study relatively homogeneous groups, we excluded from our sample unmarried individuals. We also excluded the self-employed.

This data set, which has been used in many studies of consumption (see, for instance, Attanasio and Weber (1993)), probably constitutes the best quality microdata set with consumption information and that covering the longest time period. It contains detailed information on consumption, income, and various demographic and economic variables. We report results obtained using two different definitions of consumption. The first takes as consumption expenditure on nondurable items and services, in real terms and divided by the number of adult equivalents in the household (where for the latter we use the McClements definition of adult equivalents). The second also includes expenditure on durables.

Within our model, the risk-free IBC is violated because it considers disposable income from labor and a single asset, and ignores state-contingent transfers. The latter can be interpreted as the returns to other assets with state-contingent returns or other income sources, such as government transfers. This implies that if one were to modify the definition of income to include all these state-contingent transfers, one would go back to a version of the PIH and to its implications. Therefore, one should not be able to detect excess smoothness

with this changed definition of income. Intuitively, this suggests changing the definition of income in our empirical exercise: if one considers income definitions that include transfers that are conceived to insure individual shocks, one should observe less excess smoothness when applying our tests. This, however, is not a formal statement, as our results are silent about the decentralization of the constrained efficient allocations and, empirically, we do not observe any individual or cohort of individuals for the entire life cycle. However, in what follows, we apply our tests using different definitions of income: first gross earnings, then gross earnings plus public transfers, and finally net earnings and transfers.

4.1. *The HRS Approach*

In constructing our first test, we follow HRS. They considered an income process y_t which is one element of the information structure available to the consumer and assumed that it admits the representation

$$(18) \quad (1 - L)y_t := \Delta y_t = \beta(L)w_t,$$

where w_t is an n -dimensional vector of orthogonal covariance stationary random variables that represent the information available to the consumer and $\beta(L)$ is a $1 \times n$ vector of polynomials in the lag operator L . Notice that this specification for the income process encompasses both models considered in Section 3. Here we adopt only in part the HRS notation and adapt it to ours. In particular, without loss of generality, we start from a representation for the income process that has already been rotated so that its first component represents the innovation for the process that generates consumption. It is useful to decompose the right-hand side of equation (18) into its first component and the remaining ones:

$$(19) \quad \Delta y_t = \beta_1(L)w_{1t} + \beta_2(L)w_{2t}.$$

The first result that HRS proved in their paper is that an intertemporal budget constraint has empirical content, in that it imposes testable restrictions on the time series behavior of income and consumption only if one has restrictions on the time series property of consumption, for example, that it follows a martingale. When this is not true, one can always find a time-series representation for consumption and disposable income that satisfies an IBC. In our model, a Euler equation is always satisfied because of the availability of the hidden asset, so that the HRS test has empirical content. Moreover, whether consumption satisfies a Euler equation can be checked empirically, which we do.¹⁹

¹⁹A short digression is in order here. Evidence of the “excess sensitivity” of consumption to income, as reported by Campbell and Mankiw (1989) and others, is usually interpreted as a vio-

The martingale restriction on consumption implies that it can be represented by

$$(20) \quad \Delta c_t = \pi w_{1t},$$

where deterministic trends have been removed from consumption and π is a scalar different from zero whenever insurance markets are not complete. The coefficient π represents the extent to which income news is reflected in consumption. Notice that equation (20) does not include lags. HRS showed that, given this structure, the net present value (NPV) condition implies some restrictions on the coefficients of equations (20) and (19). In particular, the intertemporal budget constraint implies that

$$(21) \quad \pi = \beta_1(\tilde{q}),$$

$$(22) \quad \beta_2(\tilde{q}) = 0,$$

where $\tilde{q} = q$ in the case in levels, while $\tilde{q} = \lambda q$ for the case in logs.²⁰ HRS showed that restriction (21) is testable, while restriction (22) is not, in that there exist other representations for income that are observationally equivalent to (19) for which the restriction holds by construction. The *alternative hypothesis* that $\pi < \beta_1(\tilde{q})$ is equivalent to what Campbell and Deaton (1989) and West (1988) defined as excess smoothness of consumption.

The theoretical structure we have illustrated in the previous section provides a structural interpretation of the excess smoothness and HRS tests. Notice the similarity of equations (14) and (15) to equations (19) and (20).²¹ The null considered by HRS corresponds to the Bewley model we considered in Section 2.3, which also corresponds to a special case of our model (see Section 3.1). The moral hazard model we constructed generates a specific deviation from the null: it implies $\beta_1(\tilde{q})/\pi = a$. In our context, the extent of excess smoothness represents the severity of the incentive problem.

lation of the orthogonality conditions implied by the Euler equation for consumption. Such evidence would constitute a problem for the use of the HRS test we are proposing. However, there is evidence that many of the violations reported in the literature are based on aggregate time series data and are explained by aggregation problems and/or the failure to control for the evolution of needs over the life cycle. See Attanasio and Weber (1993, 1995). Attanasio (2000) presented a critical discussion of these issues. Another reason that can explain income and consumption tracking each other is the nonseparability of consumption and leisure in the utility function, a point made early on by Heckman (1974), and on which Attanasio and Weber (1993, 1995) and Blundell, Browning, and Meghir (1994) provided some evidence.

²⁰HRS worked with quadratic utility and $q = \delta$. As we saw above, similar relations can be obtained for CARA utility and for CRRA when working in logs. In these last two cases, we allow for $q \neq \delta$.

²¹Simply set $\pi = \frac{1}{a}$, $w_{1t} = v_t$, $\beta_1(L) = \beta(L)$, and $\beta_2(L) = 0$.

An important feature of the HRS approach is that the test of the NPV restriction does not require the econometrician to identify *all* information available to the consumer. Intuitively, the test uses two facts. First, under the null, the intertemporal budget constraint with a single asset must hold whatever is the information set available to the agent. Second, under the assumption that the agent has no coarser information than the econometrician, the validity of the Euler equation implies that consumption innovations reveal part of the information available to the agent. By following the HRS strategy, we test the intertemporal budget constraint along the dimensions of information identified via the Euler equation (which is not zero as long as insurance markets are not complete).

The representation in equations (20) and (19) is derived under the assumption that preferences are time separable. A more general formulation of equation (20) is

$$(23) \quad \Delta c_t = \pi \psi(L) w_{1t},$$

where c in equation (23) represents total consumption expenditure, which enters the budget constraint, while utility is defined over the consumption services, which, in turn, are a function of current and, possibly, past expenditure. The polynomial in the lag operator $\psi(L)$ reflects these nonseparabilities or other complications, such as i.i.d. taste shocks to the instantaneous utility function. The nonseparabilities considered in HRS imply that $\psi(\tilde{\delta}) = 1$, where $\tilde{\delta} = \delta = q$ if utility is quadratic, while $\tilde{\delta} = 1$ with CARA and constant relative risk aversion (CRRA) utility. The condition $\psi(\tilde{\delta}) = 1$ imposes restrictions on the way lagged shocks enter the equation for consumption growth which imply that the NPV restriction, in this case, takes the form $\pi = \beta_1(\tilde{q})$, as in equation (21).²² We will need this extension to interpret some of our results.

We follow HRS and consider a time series representation of consumption and income as a function of two (unobservable) factors. As in HRS, this gives rise to an moving average (MA) representation of the form

$$(24) \quad \begin{aligned} \Delta c_{ht} &= \alpha^{cc}(L) v_{h1t}, \\ \Delta y_{ht} &= \alpha^{yc}(L) v_{h1t} + \alpha^{yy}(L) v_{h2t}, \end{aligned}$$

where we have added the subscript h to denote households, and we stress that the application will be on microdata and assume that the two vectors v are

²²Attanasio (2000, Section 3.6) discussed the case with preference shocks which leads to extended equations such as (23). HRS (in Section 4) and Alessie and Lusardi (1997) considered habit formation. HRS only considered the case with quadratic utility, while Alessie and Lusardi, within a simpler model, allowed for both quadratic and CARA utilities. The condition $\pi = \beta_1(\tilde{q})$ is immediate from their equations (3) and (9) for the quadratic and CARA utilities respectively. The CRRA utility case is similar as long as habit is multiplicative. Details are available upon request.

independent of each other and over time (we allow, however, correlation between v_{hit} and v_{kit} , $i = 1, 2$, $h \neq k$, to take into account aggregate shocks). The system (24), which is identified by a standard triangular assumption that consumption is only affected by the first factor, can be estimated by maximum likelihood, making some assumptions about the distribution of the relevant variables. As in HRS, it is straightforward to show that in the case of the intertemporal nonseparability implied by equation (23), the intertemporal budget constraint with a single asset can still be tested as a hypothesis on the coefficients of the system (24), provided that the process for consumption satisfies some restrictions we discuss below. In particular, if the IBC holds (and therefore there is no excess smoothness), the relevant restrictions to be tested are $\alpha^{cc}(\tilde{\delta}) = \alpha^{yc}(\tilde{q})$, while excess smoothness will imply $\alpha^{cc}(\tilde{\delta}) \leq \alpha^{yc}(\tilde{q})$, and once again, $\alpha^{cc}(\tilde{\delta})/\alpha^{yc}(\tilde{q}) = 1/a$ reflects the severity of the moral hazard problem in our model.

The presence of intertemporal nonseparabilities implies an equation like (23) for consumption changes and, in particular, the presence of lagged values of w_1 . Consumption is not a martingale anymore. However, as we mention above, the relevant Euler equation still provides enough restrictions to make the IBC testable. Moreover, some of these restrictions are testable in a system like (24), as they translate into restrictions on the α coefficients. While the hypothesis that $\psi(\tilde{\delta}) = 1$ cannot be tested because the coefficients of ψ are not identified ($\alpha^{cc}(\tilde{\delta})$ is *proportional* to $\psi(\tilde{q})$), we can identify coefficients on lagged values of v_2 in the first equation of the system (24) and test the hypothesis that they are zero. The alternative that they are different from zero corresponds to the standard excess sensitivity tests, which have been applied many times in the literature since Hall (1978). Intuitively, a coefficient on lagged values of v_2 in the consumption equation would imply that consumption is excessively sensitive to expected changes in income to be consistent with the PIH. We implement these tests on our data below. In our context, these tests are important because, as stressed by HRS, if consumption does not satisfy the restrictions implied by the Euler equation, we cannot meaningfully test the intertemporal budget constraint.

To implement the estimation of system (24) on our data, some important modifications of the standard procedure used by HRS are necessary. In particular, we need to take into account that we are using household-level data and that we do not have longitudinal data. To address these two problems, we use the approach recently developed by Attanasio and Borella (2006).

Because our data do not have a longitudinal dimension, we do not observe the quantities on the left-hand side of equations (24). As mentioned above, we overcome this problem by using synthetic cohort techniques (see Deaton (1985)). In particular, given groups of fixed membership, indexed by g , and an individual variable z_{gt}^h , we can state, without loss of generality,

$$z_{gt}^h = \bar{z}_{gt} + \eta_{gt}^h,$$

where the first term on the right-hand side defines the population group mean. We do not observe $\bar{z}_{gt}$, but we can obtain a consistent estimate $\tilde{z}_{ct}$ of it from our sample. This differs from the population mean by an error whose variance can be consistently estimated given the within-cell variability and cell size (see Deaton (1985)). The presence of this measurement error in the levels induces an additional MA(1) component in the time series behavior of the changes in the variables of interest. The variability of this component has to be taken into account when estimating the parameters of the model. We do so by assuming that the information on within-cell variability provides an exact measurement of the variance of this component. Given the sample sizes involved, this assumption is not very strong. Given the known values for the variance–covariance matrix of the sampling error component, the likelihood function of the MA system in (24) can be computed using the Kalman filter (for details see Attanasio and Borella (2006)).

Given these considerations, aggregating the household-level equations (24) at the group level and assuming that the degree of the polynomials in the lag operator is 2, the system that we estimate can be written as

$$(25) \quad \begin{aligned} \Delta c_{gt} &= v_{1gt} + \alpha_1^{cc} v_{1gt-1} + \alpha_2^{cc} v_{1gt-2}, \\ \Delta y_{gt} &= \alpha_0^{yc} v_{1gt} + \alpha_1^{yc} v_{1gt-1} + \alpha_2^{yc} v_{1gt-2} + v_{2gt} + \alpha_1^{yy} v_{2gt-1} + \alpha_2^{yy} v_{2gt-2}, \end{aligned}$$

where we have normalized the coefficient of the contemporaneous first factor in the consumption equation and of the second factor in the income equation to be 1. In addition to estimating the coefficients in the above system, we also test the hypothesis that coefficients on the lagged values of v_{2gt} do not enter the consumption equation.

4.1.1. Results in Levels

In Tables I and II, we report the estimates we obtain estimating the MA system (25) by maximum likelihood. As we allow the variance–covariance matrix in the system to be cohort specific, we limit the estimation to cohorts that are observed over a long time period. This means using balanced pseudopanel with two cohorts: that born in the 1940s and that born in the 1950s. We experimented with several specifications that differed in terms of the number of lags considered in the system. The most general specification included up to eight lags in both the consumption and income equation. However, no coefficient beyond lag 2 was either individually or jointly significant. In the tables, therefore, we focus on the specification with two lags.

Table I uses as a definition of consumption the expenditure on nondurables and services. We use three different definitions of income. The first is gross earnings, the second is gross earnings plus all public transfers (the most important of which are unemployment insurance and housing benefits—benefits awarded to the needy to cover housing expenses), and the third is net earnings

TABLE I
NONDURABLE CONSUMPTION^a

	Gross Earnings	Gross Earnings	Gross Earnings + Benefits	Gross Earnings + Benefits	Net Earnings + Benefits	Net Earnings + Benefits
<i>Income Equation</i>						
Own Shock						
$ayy(t)$	1	1	1	1	1	1
	—	—	—	—	—	—
$ayy(t - 1)$	0.505	—	0.333	—	0.376	—
	(20.735)	—	(25.683)	—	(18.240)	—
$ayy(t - 2)$	−0.672	—	−0.748	—	−0.507	—
	(12.252)	—	(18.183)	—	(10.247)	—
Income/Consumption Shock ^b						
$ayc(t)$	1.159	1.171	1.073	1.064	0.771	0.793
	(0.346)	(0.317)	(0.329)	(0.317)	(0.253)	(0.243)
$ayc(t - 1)$	−1.140	−1.148	−0.887	−0.894	−0.602	−0.587
	(0.454)	(0.415)	(0.492)	(0.459)	(0.342)	(0.323)
$ayc(t - 2)$	0.992	0.995	0.827	0.844	0.619	0.568
	(0.370)	(0.331)	(0.356)	(0.311)	(0.257)	(0.227)
<i>Consumption Equation</i>						
Income/Consumption Shock ^b						
$acc(t)$	1	1	1	1	1	1
	—	—	—	—	—	—
$acc(t - 1)$	−0.577	−0.493	−0.604	−0.491	−0.612	−0.499
	(0.196)	(0.114)	(0.193)	(0.116)	(0.191)	(0.111)
$acc(t - 2)$	0.084	—	0.116	—	0.118	—
	(0.201)	—	(0.191)	—	(0.193)	—
$\log L$	−773.3	−773.9	−759.5	−760.6	−685.0	−685.5
$Excess\ smoothness^c$	−0.491	−0.499	−0.489	−0.493	−0.273	−0.263
SE	(0.171)	(0.165)	(0.160)	(0.153)	(0.132)	(0.128)

(Continues)

TABLE I—Continued

	Gross Earnings	Gross Earnings	Gross Earnings + Benefits	Gross Earnings + Benefits	Net Earnings + Benefits	Net Earnings + Benefits
<i>Excess sensitivity^d</i>						
log L unrestricted model	−771.87	−773.4	−759.45	−758.9	−684.64	−684.4
LR	2.9	0.94	0.1	3.38	0.68	2.22
p -Value	0.235	0.625	0.951	0.185	0.712	0.330
<i>Comparison with 4 lags model</i>						
log L 4 lags model	−771.5	−773.6	−756.3	−760.4	−681.7	−685.3
LR	3.7	0.54	6.34	0.42	6.52	0.34
p -Value	0.717	0.763	0.386	0.811	0.368	0.844

^aAll data are in (first diff of) levels. Standard Errors (SE) are given in parentheses.

^bIncome/consumption shock is the shock that enters both the income and the consumption equation.

^cExcess smoothness tests are computed as $\text{sum}(\text{acc}(t - L)) - \text{sum}(\text{ayy}(t - L)) = 0$, with $L = 0, \dots, 4$; interest rate = 0.01.

^dExcess sensitivity test computed as LR test (restrictions: $\text{acy}(t - 1) = \text{acy}(t - 2) = 0$).

TABLE II
TOTAL CONSUMPTION EXPENDITURE^a

	Gross Earnings	Gross Earnings	Gross Earnings + Benefits	Gross Earnings + Benefits	Net Earnings + Benefits	Net Earnings + Benefits
<i>Income Equation</i>						
Own Shock						
ayy(<i>t</i>)	1	1	1	1	1	1
	—	—	—	—	—	—
ayy(<i>t</i> − 1)	51.615 (1535.290)	—	0.400 (9.124)	—	0.321 (28.027)	—
ayy(<i>t</i> − 2)	−36.884 (1072.450)	—	−0.968 (7.222)	—	−0.765 (20.287)	—
Income/Consumption Shock						
ayc(<i>t</i>)	0.596 (0.427)	0.586 (0.351)	0.730 (0.383)	0.497 (0.346)	0.683 (0.236)	0.662 (0.205)
ayc(<i>t</i> − 1)	−0.341 (0.638)	−0.673 (0.473)	−0.472 (0.572)	−0.533 (0.477)	−0.277 (0.358)	−0.273 (0.304)
ayc(<i>t</i> − 2)	0.572 (0.451)	0.981 (0.467)	0.561 (0.430)	0.910 (0.451)	0.330 (0.280)	0.363 (0.238)
<i>Consumption Equation</i>						
Income/Consumption Shock						
acc(<i>t</i>)	1	1	1	1	1	1
	—	—	—	—	—	—
acc(<i>t</i> − 1)	−0.349 (0.406)	−0.346 (0.208)	−0.386 (0.366)	−0.345 (0.202)	−0.372 (0.367)	−0.395 (0.172)
acc(<i>t</i> − 2)	−0.069 (0.358)	—	−0.022 (0.330)	—	−0.031 (0.319)	—
log <i>L</i>	−885.2	−886.0	−869.4	−870.5	−788.0	−788.9
Excess smoothness ^b	−0.233	−0.224	−0.217	−0.203	−0.131	−0.174
SE	(0.128)	(0.144)	(0.116)	(0.113)	(0.099)	(0.128)

(Continues)

TABLE II—*Continued*

	Gross Earnings	Gross Earnings	Gross Earnings + Benefits	Gross Earnings + Benefits	Net Earnings + Benefits	Net Earnings + Benefits
<i>Excess sensitivity^c</i>						
log L unrestricted model	−882.9	−885.3	−869.25	−868.7	−786.8	−788.8
LR	4.64	1.36	0.36	3.64	2.4	0.1
p -Value	0.098	0.507	0.835	0.162	0.301	0.951
<i>Comparison with 4 lags model</i>						
log L 4 lags model	−882.3	−885.4	−867.8	−869.6	−786.2	−788.3
LR	5.78	1.24	3.26	1.82	3.54	1.1
	0.448	0.538	0.776	0.403	0.739	0.577

^aAll data are in (first difference of) levels. Standard errors (SE) are given in parentheses.

^bExcess smoothness test computed as $\text{sum}(\text{axx}(t - L)) - \text{sum}(\text{ayx}(t - L)) = 0$, with $L = 0, \dots, 4$; interest rate = 0.01.

^cExcess sensitivity test computed as LR test (restrictions: $\text{acy}(t - 1) = \text{acy}(t - 2) = 0$).

plus benefits. For each of the three definitions, we report two specifications: one with two lags in each of the two equations and one where the insignificant coefficients are restricted to zero.

Several interesting elements come out of the table. First, the dynamics of income are richer than those of consumption. However, and perhaps surprisingly, the coefficients on the lags of v_{2gt} are not statistically significant and are constrained to zero in the third, fifth, and seventh columns. In the consumption equation, the coefficient on the first lag of v_{1gt} is consistently significant and attracts a negative sign. As discussed above, this could be a sign of intertemporal nonseparability of preference, maybe induced by some elements of nondurable consumption having some durability at the quarterly frequency.

To implement the HRS test of the IBC, it is necessary that the marginal utility of consumption satisfies a martingale property. In Table I, such an assumption is imposed, in that lagged income shocks do not enter the consumption equation. This hypothesis, however, can be tested. If we do not impose the restriction, the estimated coefficients are small in size and never significantly different from zero, either individually or jointly. In the table, for each specification, we report the value of the likelihood ratio (LR) test that the coefficients on the first and second lag are jointly zero and its corresponding p -value. This is an important result as it corresponds to a nonviolation of the excess sensitivity test.

The test of the intertemporal budget constraint, which is parameterized as $\pi\psi(\tilde{\delta}) - \beta_1(\tilde{q})$, clearly shows the presence of excess smoothness. Interestingly, such evidence is stronger for gross earnings. The value of the test does not change much when we add benefits to gross earnings (as in the fourth and fifth columns). However, when we consider net earnings plus benefits, the value of the test is greatly reduced in absolute value (moving from -0.49 to -0.26), although still statistically different from zero. Therefore, when we use a definition of income that includes an important smoothing mechanism, we find much less evidence of consumption excess smoothness *relative to that income definition*.

Table II mirrors the content of Table I, with the difference that the definition of consumption we use now includes expenditure on durables. The results we obtain are, in many ways, similar to those of Table I. Perhaps surprisingly, the coefficient on lagged v_{1gt} in the consumption equation is smaller in absolute value than in Table I and for two of the three income definitions is not statistically different from zero. The test of excess sensitivity, as in Table I, does not reject the null that the coefficients on lagged income shocks are zero in the consumption equation (although the p -value for the test corresponding to the second column is 0.098). The most interesting piece of evidence, however, is that the coefficient that measures excess smoothness is now considerably lower in absolute value, indicating less consumption smoothing relative to the null of the Bewley model. This is suggestive of the fact that durables might be playing an important role in the absorption of shocks, as speculated, for instance, by

Browning and Crossley (2009). However, when we consider different income definitions, the evidence is consistent with that reported in Table I, in that consumption exhibits much less excess smoothness relative to net earnings than to gross earnings.

4.1.2. *Results in Logs*

When reestimating the system using the specification in logs, we try to use the same sample used in the specification in levels. However, as we aggregate the nonlinear relationship (i.e., we take the group average of logs), we are forced to drop observations that have zero or negative income. Apart from this, the sample is the same. We report our estimates in Table III. Given the evidence on the dynamics of consumption discussed above, we only report the results for total consumption, which includes expenditure on durables. Results for nondurables and services are available upon request. Our estimates of a are obtained under the assumption that $\lambda = 1$, which corresponds to log utility, and $\delta = q$. If $q > \delta$ and $\gamma > 1$, λ would be less than unity. We have explored values for λ below 1, obtaining similar results.

As with Tables I and II, we test the hypothesis that lagged coefficients of the income shock do not appear significantly in the consumption equation. In the case of the log specification, we stress, as mentioned in Section 3.3, that one could get a rejection either because of binding liquidity constraints or because the second moments in equation (16) might depend on lagged variables. While this is possible, empirically, this excess sensitivity test does not reject the null at standard significance values.²³

As with the results in levels, we do find evidence of excess smoothness. The drop in the size of the excess-smoothness parameter when we move to definitions of income that include some smoothing mechanisms is even more dramatic than in Table II. In the last column, corresponding to net earnings plus benefits, the excess-smoothness parameter, while still negative, is not significantly different from zero. This result suggests that most of the insurance available over and above self-insurance comes from public transfers and taxation.

4.2. *The Evolution of Cross-Sectional Variances*

The empirical implications of the model we have stressed so far focus on the means of consumption and income. Because of the lack of longitudinal data at the individual level, we were forced to use synthetic panel data. This implies that purely idiosyncratic shocks are averaged away and the excess-smoothness test identifies risk sharing across cohorts. To complement this evidence, it can,

²³The issue of the estimation of log-linearized Euler equations like equation (16) is discussed in Attanasio and Low (2004). The results we find are consistent with the findings of their simulations.

TABLE III
TOTAL CONSUMPTION EXPENDITURE: LOG SPECIFICATION^a

	Gross Earnings	Gross Earnings	Gross Earnings + Benefits	Gross Earnings + Benefits	Net Earnings + Benefits	Net Earnings + Benefits
<i>Income Equation</i>						
Own shock						
ayy(t)	1	1	1	1	1	1
	—	—	—	—	—	—
ayy($t - 1$)	0.332 (14.452)	—	0.181 (4.936)	—	0.159 (1.982)	—
ayy($t - 2$)	-0.748 (10.305)	—	-0.779 (3.957)	—	-0.595 (1.426)	—
Income/Consumption Shock						
ayc(t)	0.889 (0.200)	0.816 (0.155)	0.717 (0.162)	0.748 (0.146)	0.799 (0.191)	0.570 (0.187)
ayc($t - 1$)	-1.102 (0.249)	-1.250 (0.206)	-0.685 (0.216)	-0.836 (0.177)	-0.523 (0.253)	-0.222 (0.211)
ayc($t - 2$)	0.393 (0.229)	0.660 (0.199)	0.094 (0.191)	0.223 (0.139)	-0.120 (0.214)	0.086 (0.152)
<i>Consumption Equation</i>						
Income/Consumption Shock						
acc(t)	1	1	1	1	1	1
	—	—	—	—	—	—
acc($t - 1$)	-0.601 (0.243)	-1.011 (0.008)	-0.647 (0.279)	-1.011 (0.008)	-0.578 (0.324)	-0.619 (0.125)
acc($t - 2$)	-0.411 (0.247)	—	-0.368 (0.282)	—	-0.444 (0.330)	—
log L	100.9	97.2	153.3	148.6	173.8	168.2
LR		7.4		9.38		11.16
p -Value		0.060		0.025		0.011

(Continues)

TABLE III—*Continued*

	Gross Earnings	Gross Earnings	Gross Earnings + Benefits	Gross Earnings + Benefits	Net Earnings + Benefits	Net Earnings + Benefits
<i>Excess Smoothness</i> ^b	−0.181	−0.226	−0.133	−0.141	−0.170	−0.048
<i>SE</i>	(0.100)	(0.079)	(0.050)	(0.053)	(0.066)	(0.106)
<i>Excess Sensitivity</i> ^c						
log <i>L</i> unrestricted model	101.9	97.9	153.5	149.9	174.4	168.6
LR	2.0	1.4	0.4	2.58	1.24	0.8
<i>p</i> -Value	0.368	0.497	0.819	0.275	0.538	0.670

^aAll data are in (first difference of) levels. SE given in parentheses.

^bExcess smoothness test computed as $\text{sum}(\text{axx}(t - L)) - \text{sum}(\text{ayx}(t - L)) = 0$, with $L = 0, \dots, 4$; interest rate = 0.01.

^cExcess sensitivity test computed as LR test (restrictions: $\text{acy}(t - 1) = \text{acy}(t - 2) = 0$).

therefore, be useful to consider the implications of the theory for the cross-sectional variances of income and consumption. The test we develop here focuses on risk sharing *within* cohorts. For this purpose, the closed-form solution derived in Appendix B for the model with two independent shocks, one of which is assumed to be temporary and one permanent, is particularly useful. In this case, we have

$$(26) \quad \ln c_t^i = \frac{1}{a^p} \ln x_t^i + \frac{(1-\lambda q)}{a^T} \sum_{s=0}^{t-1} \ln \xi_{t-s}^i \\ + t \frac{\gamma}{2} \left[\left(\frac{1}{a^p} \right)^2 \sigma_{v^p}^2 + \left(\frac{1-\lambda q}{a^T} \right)^2 \sigma_{v^T}^2 \right] - t \frac{\ln \frac{q}{\delta}}{\gamma} + \mu^i + z_t.$$

The term μ^i allows for ex ante heterogeneity, which could capture distributional issues, the initial level of assets of individual i , or fixed effects that can be observable to the firm but may be unobservable to the econometrician. The term z_t allows for aggregate shocks, which are assumed to be orthogonal to individual shocks and included in the information set of all agents in the economy. If we compute the cross-sectional variance at time t of both sides of equation (26), we have

$$(27) \quad \text{Var}(\ln c_t^i) = \left(\frac{1}{a^p} \right)^2 \text{Var}(\ln x_t^i) \\ + \left(\frac{1-\lambda q}{a^T} \right)^2 \text{Var} \left(\sum_{s=0}^{t-1} \ln \xi_{t-s}^i \right) + \text{Var}(\mu^i) \\ + 2 \left[\frac{\text{Cov}(\ln x_t^i, \mu^i)}{a^p} + \frac{(1-\lambda q)}{a^T} \sum_s \text{Cov}(\ln \xi_{t-s}^i, \mu^i) \right],$$

where we have used the fact that $\text{Cov}(\ln x_t, \ln \xi_{t-s}) = 0$ for all s .²⁴ We start by assuming that both $\text{Cov}(\ln \xi_{t-s}^i, \mu^i)$ and $\text{Cov}(\ln x_t^i, \mu^i)$ are time invariant. The time invariance of these terms can be obtained by assuming constant μ s across agents in the same group (as in our theoretical model). If we now take the first difference of equation (27), neglecting the individual indices, we have

$$(28) \quad \Delta \text{Var}(\ln c_t) = \left(\frac{1}{a^p} \right)^2 \Delta \text{Var}(\ln x_t) + \left(\frac{1-\lambda q}{a^T} \right)^2 \text{Var}(\xi_t^i),$$

²⁴Here we are making use of the income process with two separate shocks typically assumed in the PIH literature, which we introduce in Appendix B. If we were to use the process with a single shock used in Section 3, equation (26) for log consumption would be much more complicated, as reported in Appendix B. It is easy to see, however, that the whole analysis goes through for $\text{Cov}(\ln x_t, \ln \xi_{t-s}) \neq 0$, as long as it is time constant for all $s \geq 0$.

where $\Delta \text{Var}(\ln c_t) := \text{Var}(\ln c_t) - \text{Var}(\ln c_{t-1})$ and $\Delta \text{Var}(\ln x_t)$ is similarly defined. Under our assumptions, $\Delta \text{Var}(\ln y_t) = \Delta \text{Var}(\ln x_t + \ln \xi_t) = \Delta \text{Var}(\ln x_t)$, which implies two things. First, we can replace the permanent component x_t by total income y_t in the first term of the right-hand side of equation (28). Second, without independent identification of the two components of income (x_t and ξ_t), only $a^p (= a)$ can be identified.

Notice that equation (28) allows again the identification of the structural parameter a , which reflects the severity of the moral hazard problem. As noted by Deaton and Paxson (1994), under perfect risk sharing, the cross-sectional variance of consumption is constant over time.²⁵

Under the PIH, as pointed out by Blundell and Preston (1998), the changes in the variance of consumption reflect changes in the variance of (permanent) income. Here, we consider a specific alternative to the perfect insurance hypothesis that implies that consumption variance grows, but less than the increase in the variance of permanent income.

The empirical strategy we follow here is quite different from Blundell, Pistaferri, and Preston (2008). They studied the evolution of the cross-sectional variance of consumption *growth*,²⁶ while we start from the specification for consumption *levels* in equation (26) to derive equation (28). We notice that, as we do not have longitudinal data, we cannot identify separately a^p and a^T , while Blundell and Preston (1998) did identify the proportion of permanent and transitory shocks that can be insured.

The estimation of equation (28) also requires the identification of groups. Here the group implicitly defines the participants in a risk sharing arrangement and the test identifies the amount of risk sharing within that group. As with the estimation of the HRS system, the lack of truly longitudinal data and the use of time series of cross sections imply that the estimated variances (for income and consumption) have an error component induced by the variability of the sample. This is particularly important for the changes in the variance of income on the right-hand side: the problem induced is effectively a measurement error problem which induces a bias in the estimated coefficient. However, as explained in Appendix C, it is easy to obtain an expression for the bias implied by an ordinary least squares (OLS) estimator in finite samples and to correct it.

To estimate the parameters in equation (28), we use the same sample we used for the HRS test, with the only difference being that we do not limit ourselves to the balanced pseudopanel, but use four cohorts, although the youngest and oldest are only used for part of the time period. Otherwise, the selection criteria used to form our sample are the same as above.

²⁵ Attanasio and Szekely (2004) proposed a test of perfect risk sharing based on changes in the cross-sectional variances of marginal utilities.

²⁶ This has important advantages, but it forces the use of the approximation $\text{Var}(\Delta c_t) \approx \Delta \text{Var}(c_t)$.

TABLE IV
VARIANCE BASED TEST

Individual Var.	Nondurable Consumption	Nondurable Consumption per Adult Equiv.	Total Consumption	Total Consumption per Adult Equiv.
<i>Gross earnings</i>	0.0709 0.0133	0.0376 0.0154	0.0765 0.0177	0.0547 0.0196
Implied a	3.7556 0.0484	5.1571 0.0900	3.6144 0.0610	4.2746 0.0865
<i>Gross earnings + benefits</i>	0.2357 0.0302	0.1476 0.0355	0.3019 0.0401	0.2495 0.0448
Implied a	2.0596 0.0447	2.6032 0.0747	1.8200 0.0492	2.0021 0.0634
<i>Net earnings + benefits</i>	0.2601 0.0351	0.1466 0.0413	0.3478 0.0463	0.2733 0.0519
Implied a	1.9608 0.0482	2.6121 0.0871	1.6957 0.0511	1.9129 0.0686
Number of observ.	505	505	505	505

The results are reported in Table IV. The four data columns in the table each report the slope coefficient of equation (28) and the implied a with the corresponding standard errors. The standard error of a is computed by the delta method. In the first two data columns, we use expenditure on nondurables and services as our definition of consumption, while in the last two we use total expenditure. In the third and last columns, the consumption figure is divided by the number of adult equivalents. The three panels correspond to the same three definitions of income we used for the HRS test.

The first aspect to be noted is that all the slope coefficients are positive and statistically different from zero. Moreover, consistent with the theory, they all imply a value of a greater than unity. Finally, the results are affected only minimally by the consideration of adult equivalents.

If we analyze the difference across income definitions, we find results that are consistent with the implications of the model and, by and large, with the evidence from the HRS approach. The coefficient on the changes in the variance of gross earnings is much smaller than the coefficient on the other income definitions. This is consistent with the evidence from Tables I and II, which showed more excess smoothness for this definition. Unlike in Tables I and II, however, the main difference in the size of the coefficient is between the first income definition on one side and the second and third on the other. With the HRS approach, the main difference was between the first and second on one side and the third on the other.

Finally, if we look at the differences between the definitions of consumption that include durables and those that do not, we find that the coefficients are (except for the first income definition) larger for the former than the latter. Again, this is consistent with the evidence from the HRS approach, which finds less excess smoothness when durables are included in the definition of consumption, that is, some self-insurance mechanisms seem to be at work via durables.

The consistency of the results obtained with the variance approach and those obtained with the HRS approach is remarkable because the two tests, as stressed above, focus on different aspects of risk sharing: the latter on insurance across groups and the former on insurance within groups. It is remarkable that both yield results that are in line with our model and indicate that the observed amount of risk sharing is between that predicted by a simple permanent income model and that predicted by perfect insurance markets. Comparing the magnitudes of the coefficients from the two approaches, we can obtain a measure of the different degrees of risk sharing possibilities that are available within cohorts as opposed to those available across cohorts.

5. CONCLUSIONS

In this paper, we discuss the theoretical and empirical implications of a model where perfect risk sharing is not achieved because of information problems. A specific and important feature of our model is that in addition to the standard moral hazard problem, in the economy we study, agents have hidden access to the credit market. After characterizing the equilibrium of this model, we show how it can be useful to interpret individual data on consumption and income.

Developing results in Abraham and Pavoni (2004), we show that in a competitive equilibrium of our model, agents typically obtain more insurance than in a Bewley setup. Moreover, we are able to construct examples in which we can get closed-form solutions for consumption. These results have more than an aesthetic value: in our empirical approach they allow us to give a structural interpretation to some of the empirical results in the literature and to those we obtain. In particular, in our model one *should* observe the so-called excess smoothness of consumption. Moreover, excess smoothness is distinct from the so-called excess sensitivity of income to consumption. Finally, we can map the excess-smoothness parameter into a structural parameter.

The presence of excess smoothness follows from the fact that even in the presence of moral hazard and hidden assets, in general, a competitive equilibrium is able to provide some insurance over and above what individuals achieve on their own by self-insurance. This additional insurance is what generates excess smoothness in consumption, which can then be interpreted as a violation of the intertemporal budget constraint with a single asset. The equilibrium allocations generated by our model violate the IBC with a single asset because they neglect some state-contingent transfers the agents use to share risk.

Tests of the risk-free intertemporal budget constraint become tests of market structure in our framework. For this reason, we start our empirical work using a test of excess smoothness that was first proposed as a test of the intertemporal budget constraint with a single asset by HRS. We extend their approach so that it can be applied to microdata.

Because longitudinal data on consumption are rarely available, we work with time series of cross sections and synthetic cohort data. As we are forced to aggregate the consumption and income of individuals belonging to a given year-of-birth cohort, we necessarily lose some of the variability in idiosyncratic income and the possibility of studying the amount of risk sharing of these shocks. This is one of the reasons why, in addition to the extension of the HRS test, we propose an additional test, which uses the movements in the cross-sectional variances of consumption and income to identify the same structural parameters of our model.

While related to the work of Deaton and Paxson (1994), Blundell and Preston (1998), and Blundell, Pistaferri, and Preston (2008), our approach is different in that it focuses on the variance in the *level* rather than *changes* of consumption. Moreover, as is the case for the version of the HRS test we present, we can give the coefficients we estimate a structural interpretation in terms of our moral hazard model.

While many papers, starting with Campbell and Deaton (1989), have documented the ‘excess smoothness of consumption using aggregate time-series data, the evidence based on microdata is recent and very limited. Using our two different approaches and data from the United Kingdom, we forcefully reject both perfect risk sharing and the simple Bewley economy, while we do not reject the hypothesis of no excess sensitivity of consumption to income. More generally, our results are consistent with the model with moral hazard and hidden assets we considered. Particularly suggestive is the fact that when we consider income definitions that include smoothing mechanisms, such as social assistance and net taxes, we find less evidence of excess smoothness.

Our results have obvious policy implications, as one could, in principle, be able to quantify in terms of welfare the insurance role played by the taxation system or unemployment insurance. Such computations would be immediately feasible from our analysis. We would also be able to perform accurate counterfactuals so as to evaluate the effects of a given policy. All normative issues, however, are left for future research.

APPENDIX A

A.1. *Implementing the Efficient Allocation With Income Taxes*

In this section, we show that under some conditions the equilibrium allocation (e, c, y) can be described using a transfer scheme τ which is a function of income histories y^t alone. This simplifies the analysis and allows us to describe the consumption allocation in terms of observables.

Notice first that through $y_t = \mathbf{y}_t(\theta^t)$, the y component of the equilibrium allocation generates histories of income levels y^t . Let us denote by $y^t(\theta^t) = (y_1(\theta^1), \dots, y_t(\theta^t))$ this mapping. In general, $\mathbf{y}_t(\theta^t)$ is not invertible, as it might be the case that for a positive measure of histories of shocks θ^t , we get the same history of incomes y^t . A generalization of the argument used in Kocherlakota (2005), however, shows that it suffices to assume that the optimal plan of consumption c alone is y^t -measurable. That is, that there exists a sequence of y^t -measurable functions c^* such that for all t, θ^t , we have $\mathbf{c}_t^*(y^t(\theta^t)) = \mathbf{c}_t(\theta^t)$. We now show that under fairly general conditions, the implementation idea of Kocherlakota (2005) extends to the general case with hidden savings.

Now notice that $\mathbf{y}_t$ is y^t -measurable by construction. As a consequence, from (9) is easy to see that the y^t -measurability of c implies that τ is y^t -measurable as well. From the transfer scheme τ , we can hence obtain the new y^t -measurable scheme τ^* as follows: $\tau_t^*(y^t(\theta^t)) = \tau_t(\theta^t)$. Given τ^* , let

$$\begin{aligned} & \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(c_t^*, \hat{e}_t) \middle| \hat{e} \right] \\ & := \sum_{t=1}^T \delta^{t-1} \int_{\Theta^t} u(\mathbf{c}_t^*(\hat{y}^t(\theta^t)), \hat{e}_t(\theta^t)) d\mu^t(\theta^t), \end{aligned}$$

where $\mathbf{c}_t^*(\hat{y}^t(\theta^t)) = \tau_t^*(\hat{y}^t(\theta^t)) + \mathbf{y}_t^*(\hat{y}^t(\theta^t))$ and the new mapping is induced by $\hat{e}$ as follows: $\hat{y}_t(\theta^t) = f(\theta_t, \hat{e}_t(\theta^t))$ for all t, θ^t . For any history of shocks θ^t , a plan $\hat{e}$ not only entails different effort costs, it also generates a different distribution over income histories y^t , hence on transfers and consumption. This justifies our notation for the conditional expectation.

We say that the equilibrium allocation (e, c, y) can be described with y^t -measurable transfers if the agent does not have an incentive to deviate from c^*, e^* given τ^* . The incentive constraint in this case is

$$(29) \quad \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(c_t^*, e_t^*) \middle| e^* \right] \geq \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(\hat{c}_t, \hat{e}_t) \middle| \hat{e} \right],$$

where, as usual, the deviation path of consumption $\hat{c}$ must be replicated by a plan of risk-free bonds $\hat{b}$. An important restriction in the deviations $\hat{e}$ contemplated in constraint (29) is that they are required to generate attainable histories of y s, that is, histories of y s that can happen in an equilibrium allocation. The idea is that any off-the-equilibrium value for y^t will detect a deviation with certainty. One can hence set the firm's transfers to a very low value (perhaps minus infinity) in these cases, so that the agent never has an incentive to generate such off-the-equilibrium histories.

Finally, suppose the agent chooses an effort plan $\hat{e}$ so that the realized history $\hat{y}^t$ is attainable in equilibrium. This means both that there is a reporting

strategy $\hat{\sigma}$ so that $\hat{y}^t = (y_1(\hat{\sigma}), y_2(\hat{\sigma}), \dots, y_t(\hat{\sigma}))$ and that, given a consumption plan $\hat{c}$, the utility the agent gets is $\mathbf{E}[\sum_{t=1}^T \delta^{t-1} u(\hat{c}_t, e_t) | (e, c, y); \hat{\sigma}]$, where the notation is that in the main text.²⁷ This effectively completes the proof since the incentive constraint (4) guarantees that the agent chooses the truth-telling strategy which implies the equilibrium plans e and c as optimal for him.

A.2. Proof of Proposition 1

We now show that, for the specification of preferences and production function we stated in Proposition 1, incentive compatibility fully characterizes the efficient allocation. In fact, we allow for a generic sequence of bond prices $\{q_t\}_{t=1}^{T-1}$ faced by both the agents and the firms in the economy.

To avoid the use of the taxation principle, we consider transfer schemes that depend on the revelation plan σ . This notation also is more directly related to the Bewley model of Section 2.3. Recall that $T < \infty$ and consider the last period of the program. Using the budget constraint, assuming the agent declared history θ^{T-1} so far, and has wealth b_T , after the realization of θ_T , his/her preferences over the report σ_T of θ_T can be represented as

$$u(c_T - e_T) = u(\theta_T + \tau_T(\theta^{T-1}, \sigma_T) + b_T).$$

The key aspect to notice here is that since utility depends on the declaration σ_T only through the transfer, the agent will declare the productivity level that delivers the maximal transfer whenever possible. More precisely, for any value of $e_T(\theta^{T-1}, \theta_T)$, the local incentive compatibility implies that for a sufficiently small $\varepsilon > 0$, we have

$$\begin{aligned} u(\theta_T + \tau_T(\theta^{T-1}, \theta_T) + b_T) \\ \geq u(\theta_T + \tau_T(\theta^{T-1}, \sigma_T) + b_T) \quad \text{for all } \sigma_T \in [\theta - \varepsilon, \theta + \varepsilon]. \end{aligned}$$

²⁷More in detail, notice that by definition, we have $\hat{c}_t(\theta^t) = g(\theta_t, \hat{y}_t(\theta^t))$. Since $\hat{y}_t(\theta^t)$ is attainable, it can be induced from y by a “lie,” that is, there exists a $\hat{\sigma}$ such that $\hat{y}_t(\theta^t) = y_t(\hat{\sigma}(\theta^t))$. But then $\hat{c}_t(\theta^t) = g(\theta_t, y_t(\hat{\sigma}(\theta^t)))$ and from the definition of τ^* , we have $c_t^*(\hat{y}_t(\theta^t)) = c_t^*(y_t(\hat{\sigma}(\theta^t))) = c_t(\hat{\sigma}(\theta^t))$, which implies that

$$\mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(c_t^*, \hat{c}_t) \middle| \hat{c} \right] = \mathbf{E} \left[\sum_{t=1}^T \delta^{t-1} u(c_t, e_t) \middle| (e, c, y), \hat{\sigma} \right]$$

for some $\hat{\sigma} \in \Sigma$. Finally, it is easy to see that c^* (and $b^* \equiv 0$) is incentive compatible among the deviations that solve $\hat{c}_t^\sigma(\sigma(\theta^t)) + q \hat{b}_{t+1}^\sigma(\sigma(\theta^t)) - \hat{b}_t^\sigma(\sigma(\theta^{t-1})) = c_t^*(\sigma(\theta^t))$ and $\lim_{t \rightarrow T} q^{t-1} \hat{b}_{t+1}^\sigma(\sigma(\theta^t)) \geq 0$ a.s. for all histories θ^t . A final remark: One can easily show that under the same conditions, e must also be x^t -measurable. If e is not x^t -measurable, it means that for at least two some θ^t and $\tilde{\theta}^t$, we have $e_t(\theta^t) \neq e_t(\tilde{\theta}^t)$ while $f(\theta_t, e_t(\theta^t)) = f(\theta_t, e_t(\tilde{\theta}^t))$. However, since u is decreasing in e , effort incentive compatibility (at $b_t = 0$) implies that $\tau_s(\theta^s) \neq \tau_s(\tilde{\theta}^s)$ for some $s \geq t$ with τ not y^s -measurable. This is contradiction to the fact that the equilibrium transfer scheme τ is y^t -measurable.

This condition applied to all $\theta_T \in \Theta$ implies that a transfer payment τ_T which is invariant across θ_T s is the sole incentive compatible possibility.

Now consider the problem in period $T - 1$. We now show that, given θ^{T-2} and θ_{T-1} , an incentive compatible transfer scheme must induce a constant number for the present value of transfers $\tau_{T-1}(\theta^{T-1}) + q_{T-1}\tau_T(\theta^{T-1})$ across θ_{T-1} levels. Note that in our notation we used the fact that τ is constant across θ_T . Consider a generic history θ^{T-2} and suppose that $\tau_{T-1}(\theta^{T-2}, \theta'_{T-1}) + q_{T-1}\tau_T(\theta^{T-2}, \theta'_{T-1}) < \tau_{T-1}(\theta^{T-2}, \theta''_{T-1}) + q_{T-1}\tau_T(\theta^{T-2}, \theta''_{T-1})$ for θ' sufficiently close to θ'' . This cannot be incentive compatible. The agent who receives shock θ'_{T-1} would declare θ''_{T-1} (and change effort by the difference between the two productivity levels) and adjust his/her bond plan accordingly. We just have to show that there is a bond plan so that the agent is better off by declaring θ''_{T-1} . This is easy to see since τ_T does not depend on θ_T . More precisely, if c_{T-1} , $c_T(\theta_T)$ is the consumption plan under θ'_{T-1} , then $c_{T-1} + \frac{\kappa}{1+q_{T-1}}$ and $c_T(\theta_T) + \frac{\kappa}{1+q_{T-1}}$ comprise one of the consumption paths attainable by deviating, where $\kappa > 0$ is the difference between the two net present values of transfers. A similar argument applies to the case where the inequality is reversed. In this case, the profitable deviation would be taken by the agent who receives shock θ''_{T-1} . The discounted value of transfer must hence be constant across θ_{T-1} as well, that is, $\tau_{T-1}(\theta^{T-1}) + q_{T-1}\tau_T(\theta^T)$ is θ^{T-2} -measurable. Going backward, we get that for all t , the quantity

$$\text{NPV}_t := \sum_{n=0}^{T-t} \left(\prod_{s=1}^n q_{t-1+s} \right) \tau_{t+n}(\theta^{t+n})$$

is at most θ^{t-1} -measurable since it is a constant number for all continuation histories $\theta_t, \theta_{t+1}, \dots, \theta_T$ following node θ^{t-1} . If we consider the initial period ($t = 1$), (local) incentive compatibility implies that $\text{NPV}_1(\theta^T) := \sum_{n=1}^T (\prod_{s=1}^{n-1} q_s) \tau_n$ is one number a.s. for all histories θ^T . As we argued already in the main text while discussing equation (5), another necessary condition for incentive compatibility is the agent's Euler equation. The problem of the firm facing a relaxed (local) incentive compatibility constraint hence reduces to the choice of the unique number NPV_1 and the plan of bond holdings for the agent. Since the agent and the firm face the same sequence of bond prices, it is easy to see that the firm is indifferent across all bond plans. This implies that the only number NPV_1 consistent with nonnegative profits and with maximizing the agent's utility is zero. Moreover, this number can be obtained by setting $\tau_i \equiv 0$ (equivalently, we could assume the transfer τ is set so that the agent chooses $b_i \equiv 0$). It is now obvious that $\tau_i \equiv 0$ is globally incentive compatible, implying that it must be optimal. We have hence shown that the optimal allocation corresponds to the bond economy allocation. Q.E.D.

More extensively, in all cases, the optimal contract must solve the standard Euler equation at each node and the condition

$$\begin{aligned}\text{NPV}_1(\theta^T) &:= \sum_{t=1}^T \left(\prod_{s=1}^{t-1} q_s \right) \tau_t^*(\theta^t) \\ &= \sum_{t=1}^T \left(\prod_{s=1}^{t-1} q_s \right) (\mathbf{c}_t^*(\theta^t) - \mathbf{y}_t^*(\theta^t)) = 0 \quad \text{a.s. for all } \theta^T.\end{aligned}$$

But these are precisely the two defining properties for the self-insurance allocation: first, it must satisfy the Euler equation; second, it must satisfy the intertemporal budget constraint with the risk free bond, that is, the period zero net present value must be zero for all θ^T . Since the Euler equation (5) is always satisfied in our allocation, the only way to obtain a different allocation is that the transfer scheme τ be permitted to violate the agent's period zero self-insurance intertemporal budget constraint for some history θ^T . The whole point of Proposition 1 is to demonstrate that this cannot be the case.

A final remark: Although the proof uses finite time, by adapting the proof of Proposition 7 in Cole and Kocherlakota (2001), we can show the result for $T = \infty$, at least as long as u is a bounded function.

REFERENCES

- ABOWD, J., AND D. CARD (1989): "On the Covariance Structure of Earnings and Hours Changes," *Econometrica*, 57, 411–445. [1041]
- ABRAHAM, A., AND N. PAVONI (2004): "Efficient Allocations With Moral Hazard and Hidden Borrowing and Lending," Working Paper 04-05, Duke University, Department of Economics. [1031,1037,1039,1061]
- (2005): "The Efficient Allocation of Consumption Under Moral Hazard and Hidden Access to the Credit Market," *Journal of the European Economic Association*, 2005, 370–381. [1031]
- (2008): "Efficient Allocations With Moral Hazard and Hidden Borrowing and Lending: A Recursive Formulation," *Review of Economic Dynamics*, 11, 781–803. [1031]
- ALESSIE, R., AND A. LUSARDI (1997): "Consumption, Savings, and Habit Formation," *Economic Letters*, 55, 103–108. [1047]
- ALLEN, F. (1985): "Repeated Principal-Agent Relationship With Lending and Borrowing," *Economic Letters*, 17, 27–31. [1036,1039]
- ATTANASIO, O. P. (2000): "Consumption," in *Handbook of Macroeconomics*, ed. by J. B. Taylor and M. Woodford. Amsterdam: Elsevier. [1046,1047]
- ATTANASIO, O. P., AND M. BORELLA (2006): "Stochastic Components of Individual Consumption: A Time Series Analysis of Grouped Data," Working Paper 12456, NBER. [1048,1049]
- ATTANASIO, O. P., AND H. W. LOW (2004): "Estimating Euler Equations," *Review of Economic Dynamics*, 7, 405–435. [1055]
- ATTANASIO, O. P., AND N. PAVONI (2011): "Supplement to 'Risk Sharing in Private Information Models With Asset Accumulation: Explaining the Excess Smoothness of Consumption,'" *Econometrica Supplemental Material*, 79, http://www.econometricsociety.org/ecta/Supmat/7063_extensions.pdf; [http://www.econometricsociety.org/ecta/Supmat/7063_data and programs.zip](http://www.econometricsociety.org/ecta/Supmat/7063_data_and_programs.zip). [1030]

- ATTANASIO, O. P., AND E. M. SZEKELY (2004): "Wage Shocks and Consumption Variability in Mexico During the 1990s," *Journal of Development Economics*, 73, 1–25. [1059]
- ATTANASIO, O. P., AND G. WEBER (1993): "Consumption Growth, the Interest Rate and Aggregation," *Review of Economic Studies*, 60, 631–649. [1044,1046]
- (1995): "Is Consumption Growth Consistent With Intertemporal Optimization? Evidence From the Consumer Expenditure Survey," *Journal of Political Economy*, 103, 1121–1157. [1046]
- BANKS, J., R. BLUNDELL, AND A. BRUGIAVINI (2001): "Risk Pooling, Precautionary Saving and Consumption Growth," *Review of Economic Studies*, 68, 757–779. [1043]
- BEWLEY, T. (1977): "The Permanent Income Hypothesis: A Theoretical Formulation," *Journal of Economic Theory*, 16, 252–292. [1035]
- BLUNDELL, R., AND I. PRESTON (1998): "Consumption Inequality and Income Uncertainty," *Quarterly Journal of Economics*, 113, 603–640. [1059,1062]
- BLUNDELL, R., M. BROWING, AND C. MEGHIR (1994): "Consumer Demand and the Life-Cycle Allocation of Household Expenditures," *Review of Economic Studies*, 61, 57–80. [1046]
- BLUNDELL, R., L. PISTAFERRI, AND I. PRESTON (2008): "Consumption Inequality and Partial Insurance," *American Economic Review*, 98, 1887–1921. [1041,1043,1059,1062]
- BROWNING, M., AND T. F. CROSSLEY (2009): "Shocks, Stocks and Socks: Smoothing Consumption Over a Temporary Income Loss," *Journal of the European Economic Association*, 7, 1169–1192. [1055]
- BROWNING, M., A. DEATON, AND M. IRISH (1985): "A Profitable Approach to Labor Supply and Commodity Demands over the Life-Cycle," *Econometrica*, 53, 503–544. [1044]
- CAMPBELL, J. Y., AND A. DEATON (1989): "Why Is Consumption so Smooth?" *Review of Economic Studies*, 56, 357–374. [1027-1029,1046,1062]
- CAMPBELL, J. Y., AND N. G. MANKIW (1989): "Consumption, Income, and Interest Rates: Reinterpreting the Time Series Evidence," in *NBER Macroeconomic Annual*, ed. by O. Blanchard, and S. Fischer. Cambridge: MIT Press, 185–216. [1045]
- COLE, H. L., AND N. R. KOCHERLAKOTA (2001): "Efficient Allocations With Hidden Income and Hidden Storage," *Review of Economic Studies*, 68, 523–542. [1036,1039,1066]
- CONSTANTINIDES, G. M., AND D. DUFFIE (1996): "Asset Pricing With Heterogeneous Consumers," *Journal of Political Economy*, 104, 219–240. [1042]
- DEATON, A. (1985): "Panel Data From Time Series of Cross-Sections," *Journal of Econometrics*, 30, 109–126. [1044,1048,1049]
- DEATON, A., AND C. PAXSON (1994): "Intertemporal Choice and Inequality," *Journal of Political Economy*, 102, 437–467. [1059,1062]
- FUDENBERG, D., B. HOLMSTROM, AND P. MILGROM (1990): "Short-Term Contracts and Long-Term Relationships," *Journal of Economic Theory*, 51, 1–31. [1036]
- GALI, J. (1991): "Budget Constraints and Time-Series Evidence on Consumption," *American Economic Review*, 81, 1238–1253. [1028]
- GOLOSOV, M., AND A. TSYVINSKI (2007): "Optimal Taxation With Endogenous Insurance Markets," *Quarterly Journal of Economic*, 122, 487–534. [1031,1032]
- HALL, R. E. (1978): "Stochastic Implications of the Permanent Income Hypothesis: Theory and Evidence," *Journal of Political Economy*, 96, 971–987. [1048]
- HANSEN, L. P., W. ROBERDS, AND T. J. SARGENT (1991), "Time Series Implications of Present Value Budget Balance and of Martingale Models of Consumption and Taxes," in *Rational Expectations Econometrics*, ed. by L. P. Hansen, and T. J. Sargent. Boulder: Westview, 121–161. [1027,1029]
- HECKMAN, J. J. (1974): "Life Cycle Consumption and Labor Supply: An Explanation of the Relationship Between Income and Consumption Over the Life Cycle," *American Economic Review*, 64, 188–194. [1046]
- KOCHERLAKOTA, N. R. (2005): "Zero Expected Wealth Taxes: A Mirrlees Approach to Dynamic Optimal Taxation," *Econometrica*, 74, 1587–1621. [1063]

- LIGON, E. (1998): "Risk Sharing and Information in Village Economies," *Review of Economic Studies*, 65, 847–864. [1043]
- MEGHIR, C., AND L. PISTAFERRI (2004): "Income Variance Dynamics and Heterogeneity," *Econometrica*, 72, 1–32. [1041]
- MIRRLEES, J. A. (1971): "An Exploration in the Theory of Optimum Income Taxation," *Review of Economic Studies*, 38, 175–208. [1041]
- NALEWAIK, J. J. (2006): "Current Consumption and Future Income Growth: Synthetic Panel Evidence," *Journal of Monetary Economics*, 53, 2239–2266. [1043]
- ROGERSON, W. (1985): "Repeated Moral Hazard," *Econometrica*, 53, 69–76. [1043]
- WEST, K. D. (1988): "The Insensitivity of Consumption to News About Income," *Journal of Monetary Economics*, 21, 17–34. [1028,1029,1046]

Dept. of Economics, University College London, Gower Street, London WC1E 6BT, United Kingdom and IFS and NBER; o.attanasio@ucl.ac.uk
and

Bocconi University and IGIER, via Roentgen 1, 20136 Milan, Italy, and University College London, and IFS, and CEPR; nicola.pavoni@unibocconi.it.

Manuscript received March, 2007; final revision received March, 2010.